

Learning from Temporal data

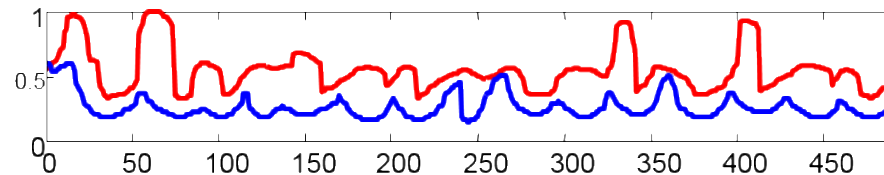
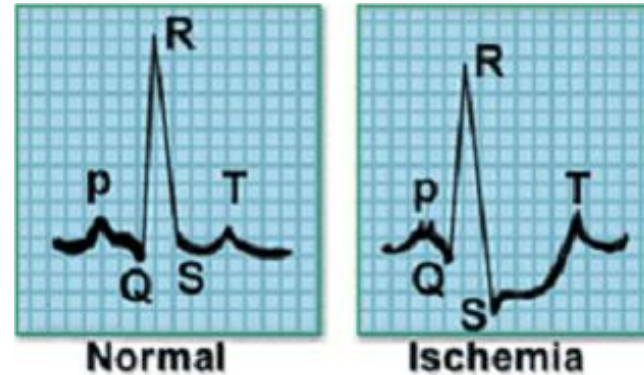
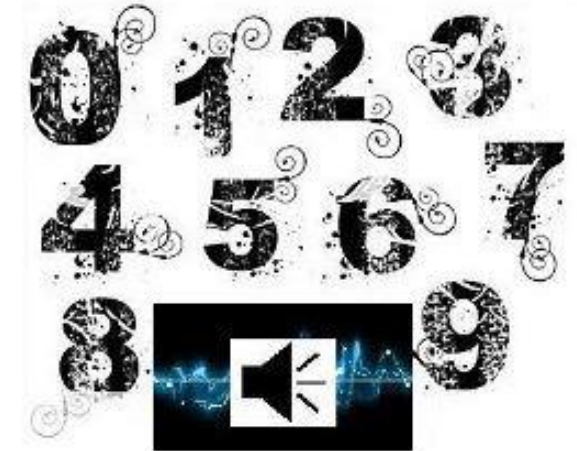
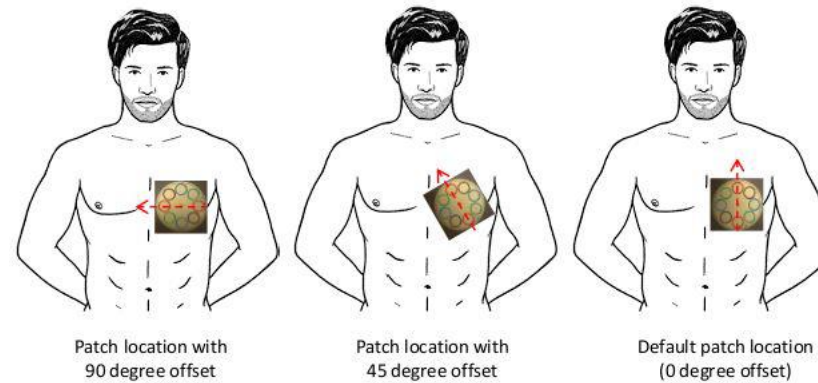
Zahraa S. Abdallah

Senior Lecturer in Machine Learning

School of Engineering Mathematics and Technology

Temporal data

Temporal data is an ordered list of observations of real valued variable (doesn't have to be time)



Why Time Series are Methodologically Different?

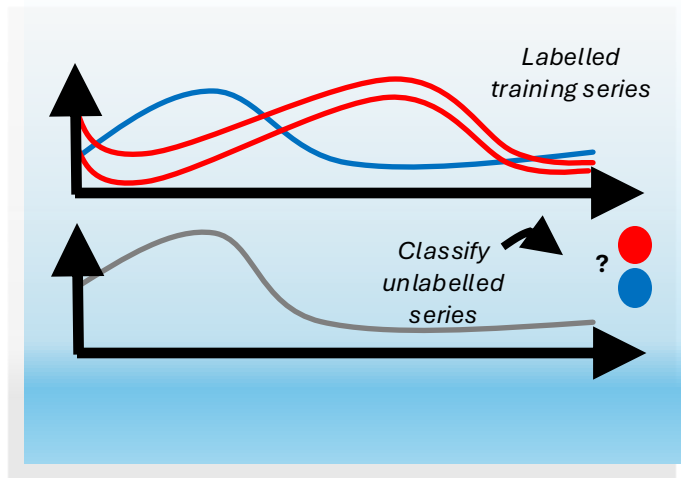
- Observations are **temporally ordered**
- Strong **dependence** between neighbouring points
- Distributions may **change over time**

Independent and identically distributed (iid) assumption does not hold.

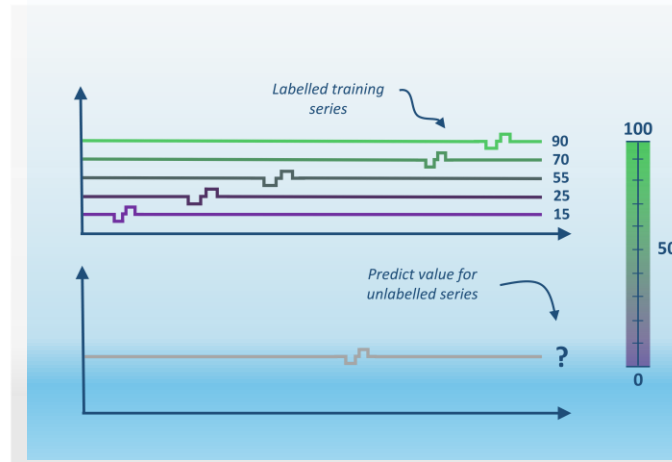


Time series (main) tasks

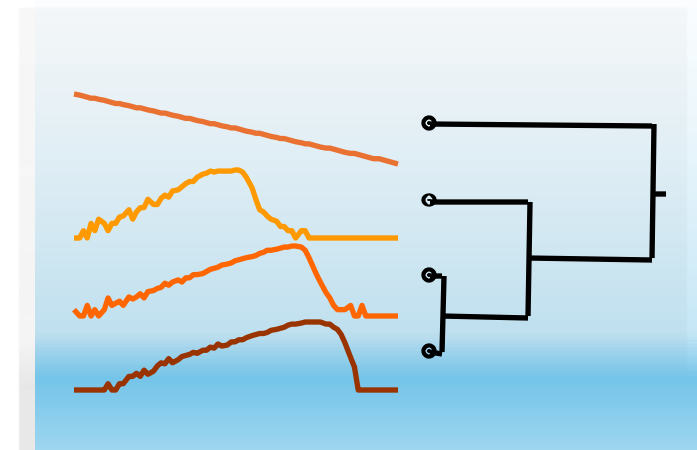
Classification



Regression



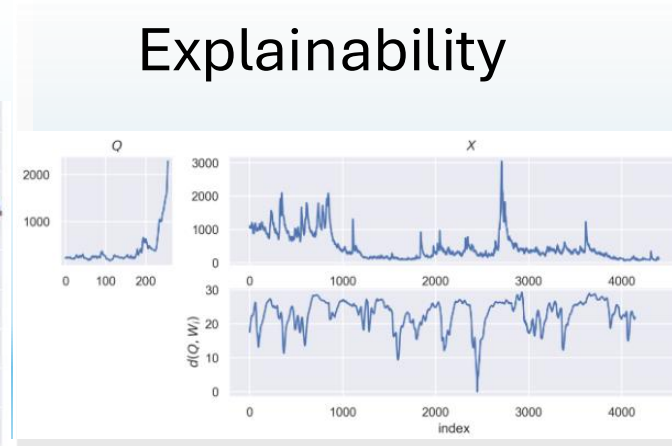
Clustering



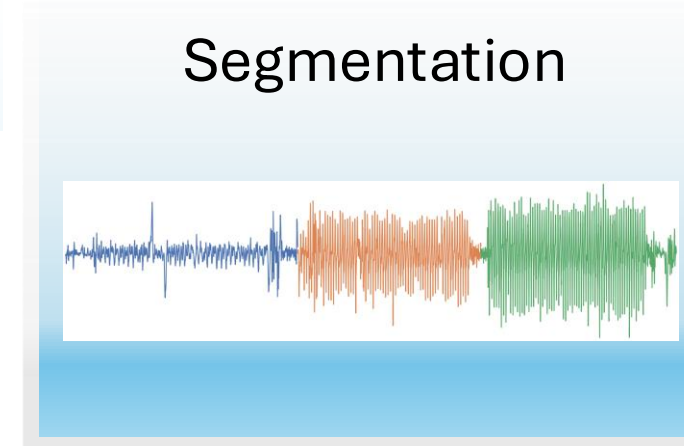
Forecasting



Explainability



Segmentation



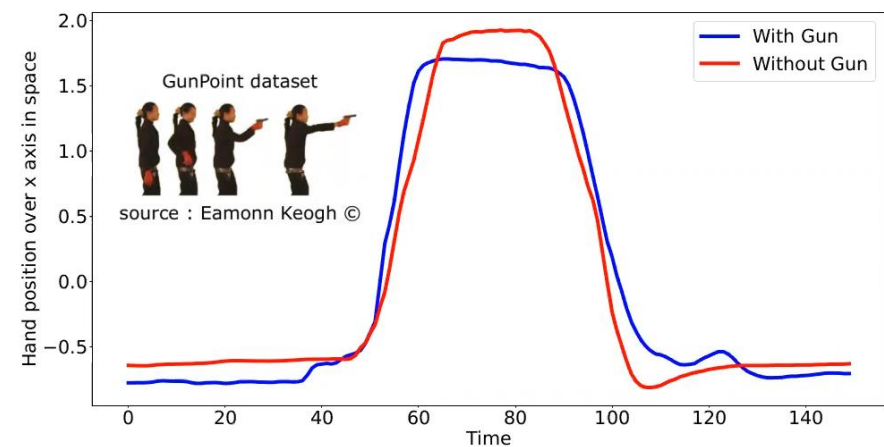
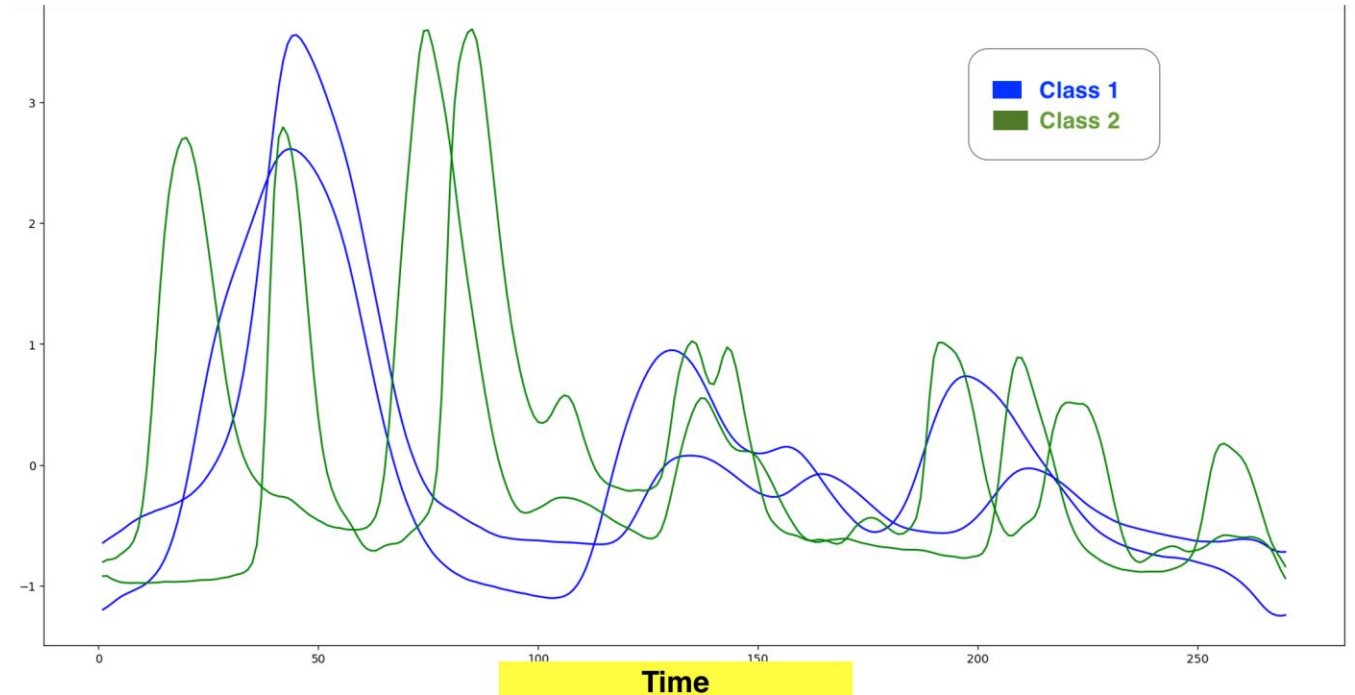
Time Series Classification

- Given a set of time series with class labels, can we train a model to accurately predict the class of new time series?

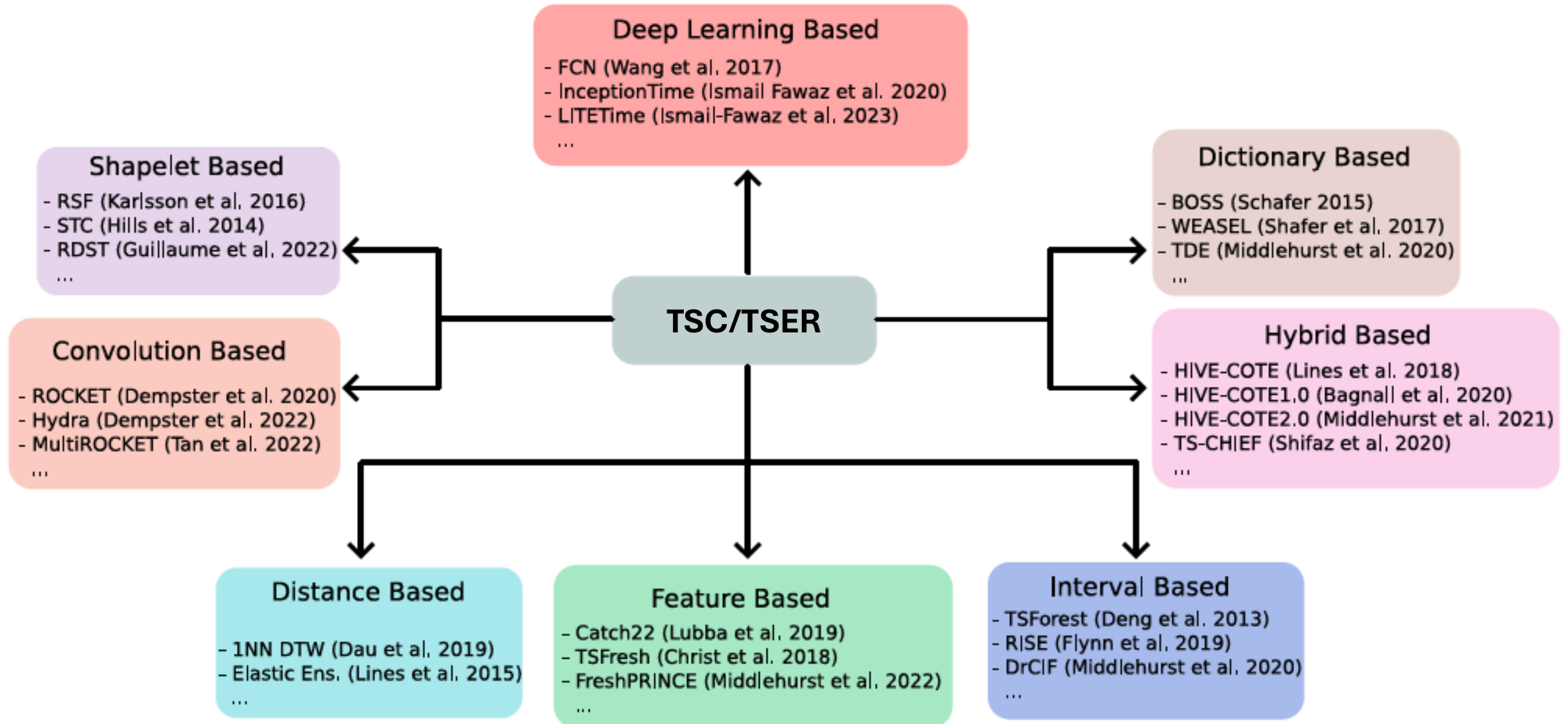
Varied-length

Irregular sampling

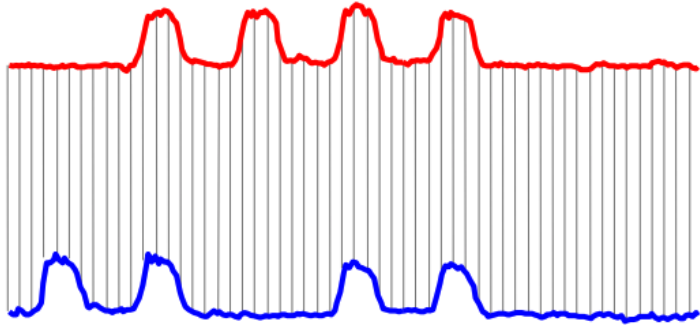
Multi-dimensions



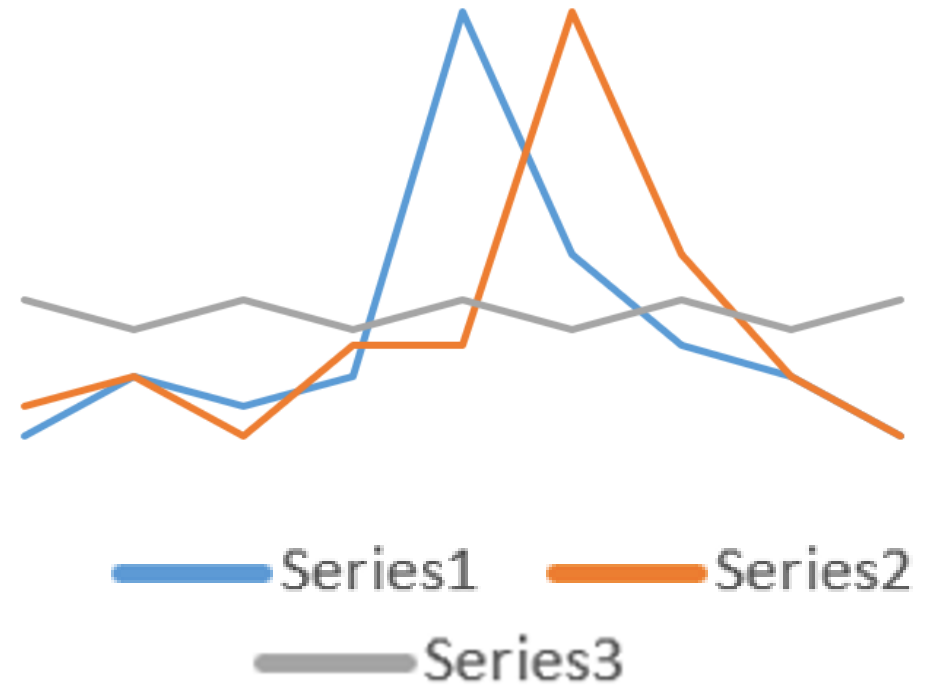
A Taxonomy of Algorithms



Distance measures



Euclidean distance requires one to
one alignment.

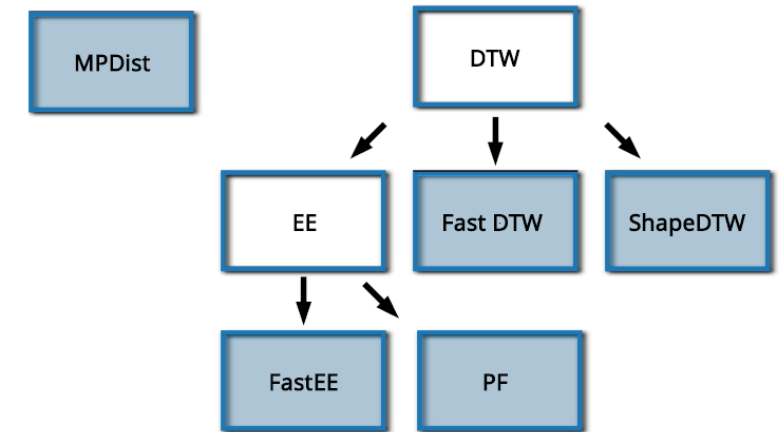
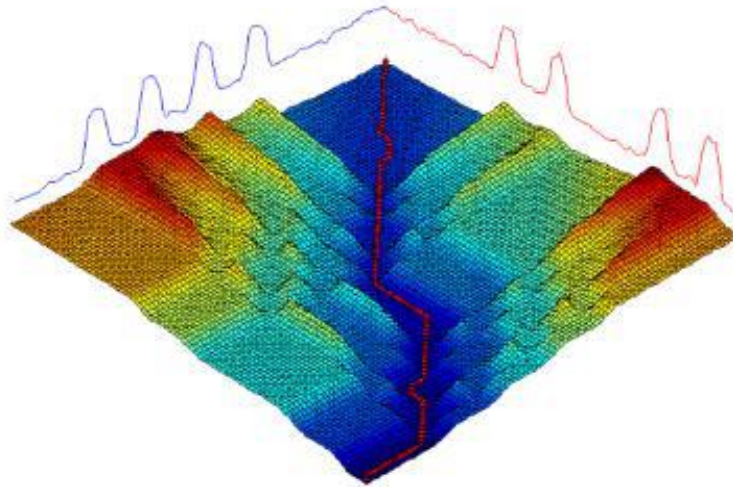
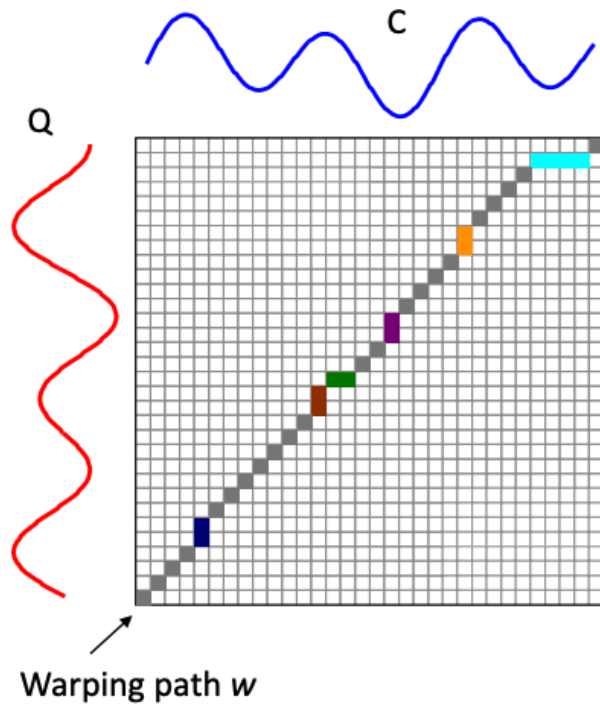


Is series 1 more like series 2 or series 3?

$$\text{dist}(s1, s2) = 14.03$$

$$\text{dist}(s1, s3) = 12.58$$

Dynamic Time Warping (DTW)



Algorithm 1 DTW ($\mathbf{a}, \mathbf{b}$, (both series of length m), w (window proportion, default value $w \leftarrow 1$), M (pointwise distance matrix))

1: Let C be an $(m+1) \times (m+1)$ matrix initialised to zero, indexed from zero.

2: **for** $i \leftarrow 1$ to m **do**

3: **for** $j \leftarrow 1$ to m **do**

4: **if** $|i - j| < w \cdot m$ **then**

5: $C_{i,j} \leftarrow M_{i,j} + \min(C_{i-1,j-1}, C_{i-1,j}, C_{i,j-1})$

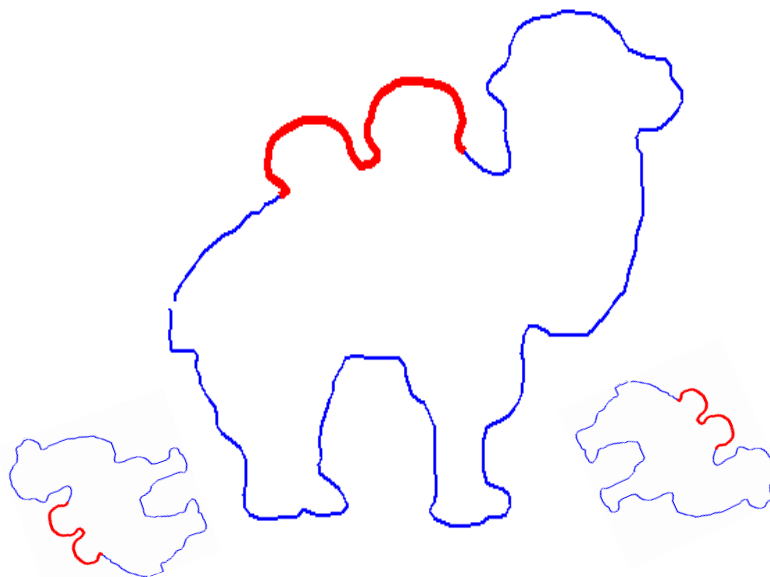
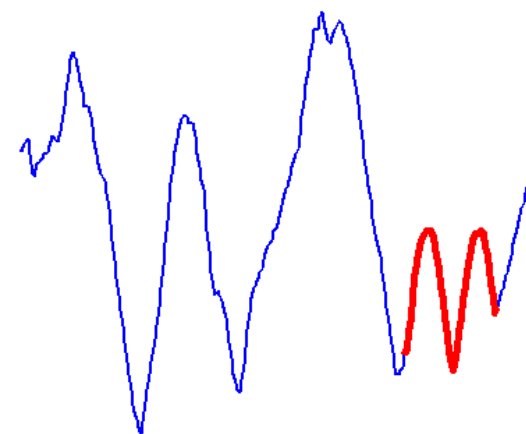
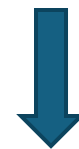
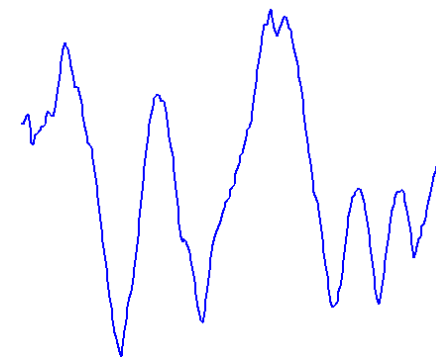
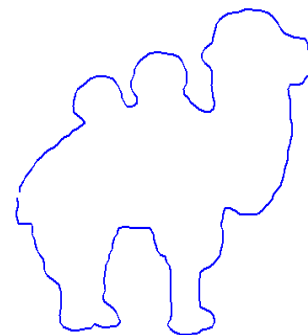
return $C_{m,m}$

DTW has no explicit penalty for moving off the diagonal.

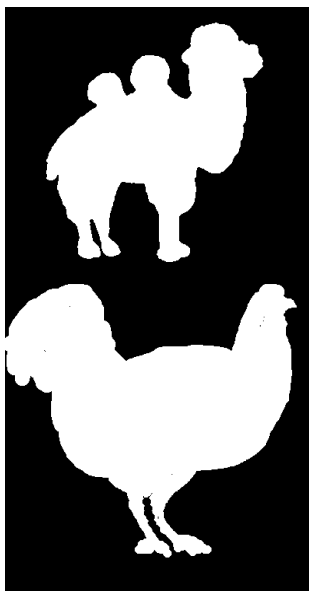
A more elastic measure is DTW, when Nonlinear alignments are possible. KNN with DTW is commonly used as a benchmark for evaluating TSC.

Shapelet-based Algorithms

Shapelets are
discriminatory phase
independent subseries



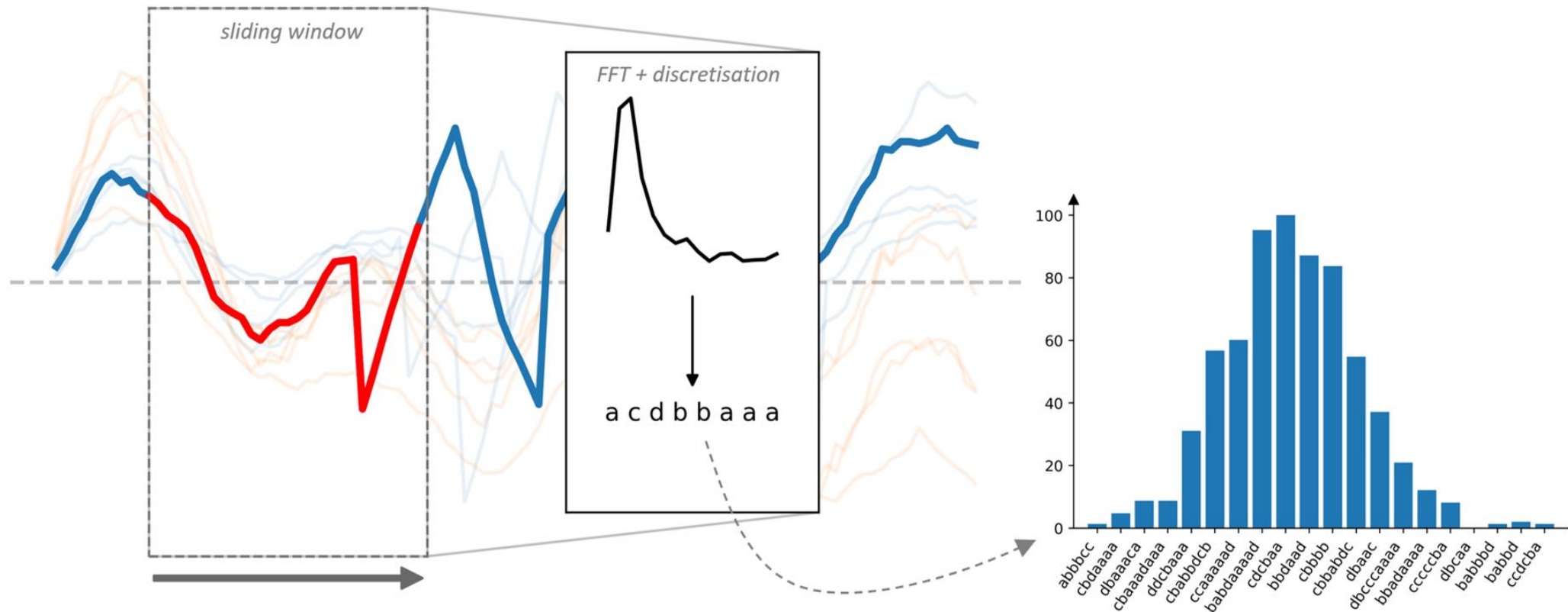
Camel or Chicken?



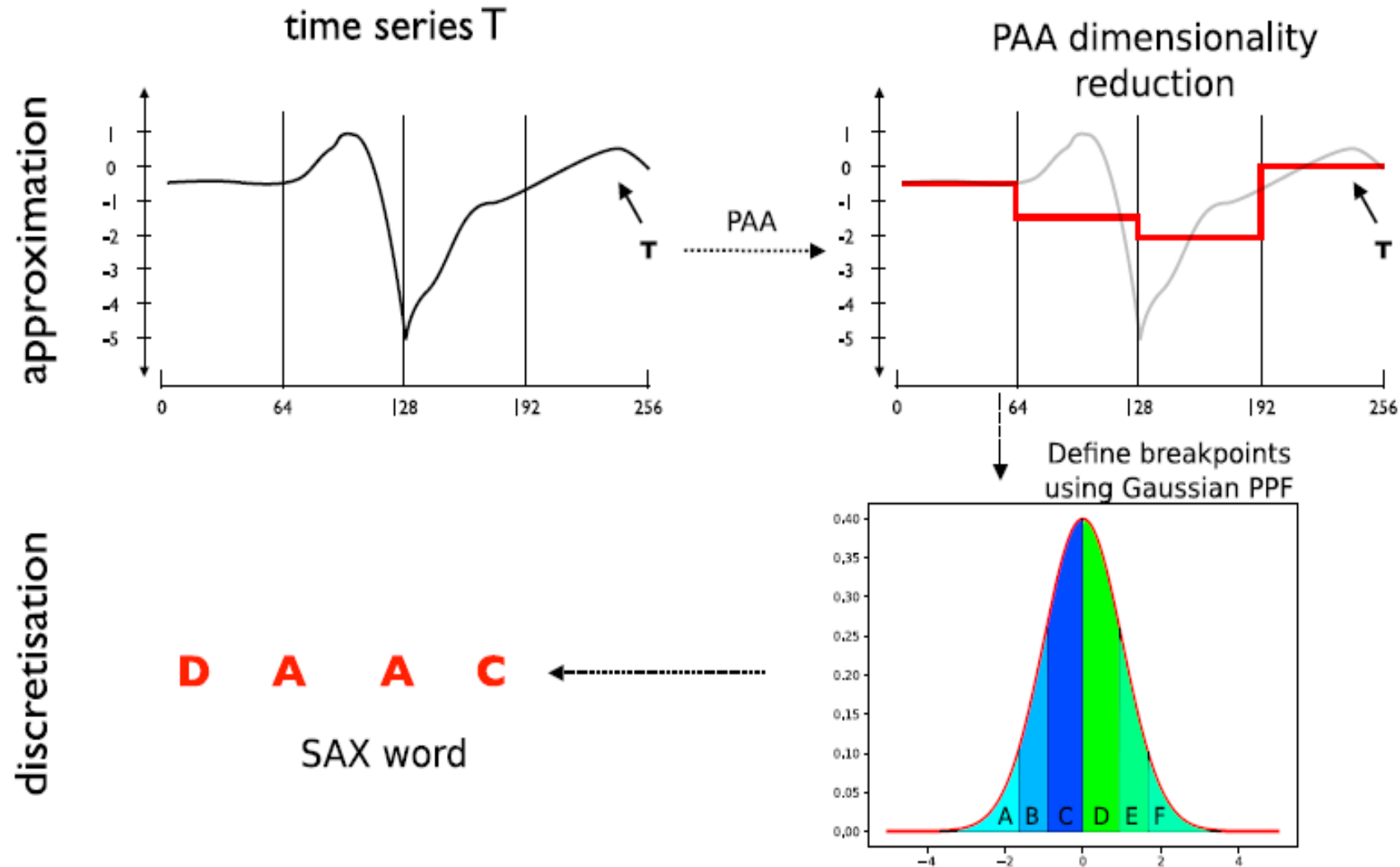
Dictionary-based Algorithms

Dictionary methods use the bag-of-words model commonly found in areas such as computer vision.

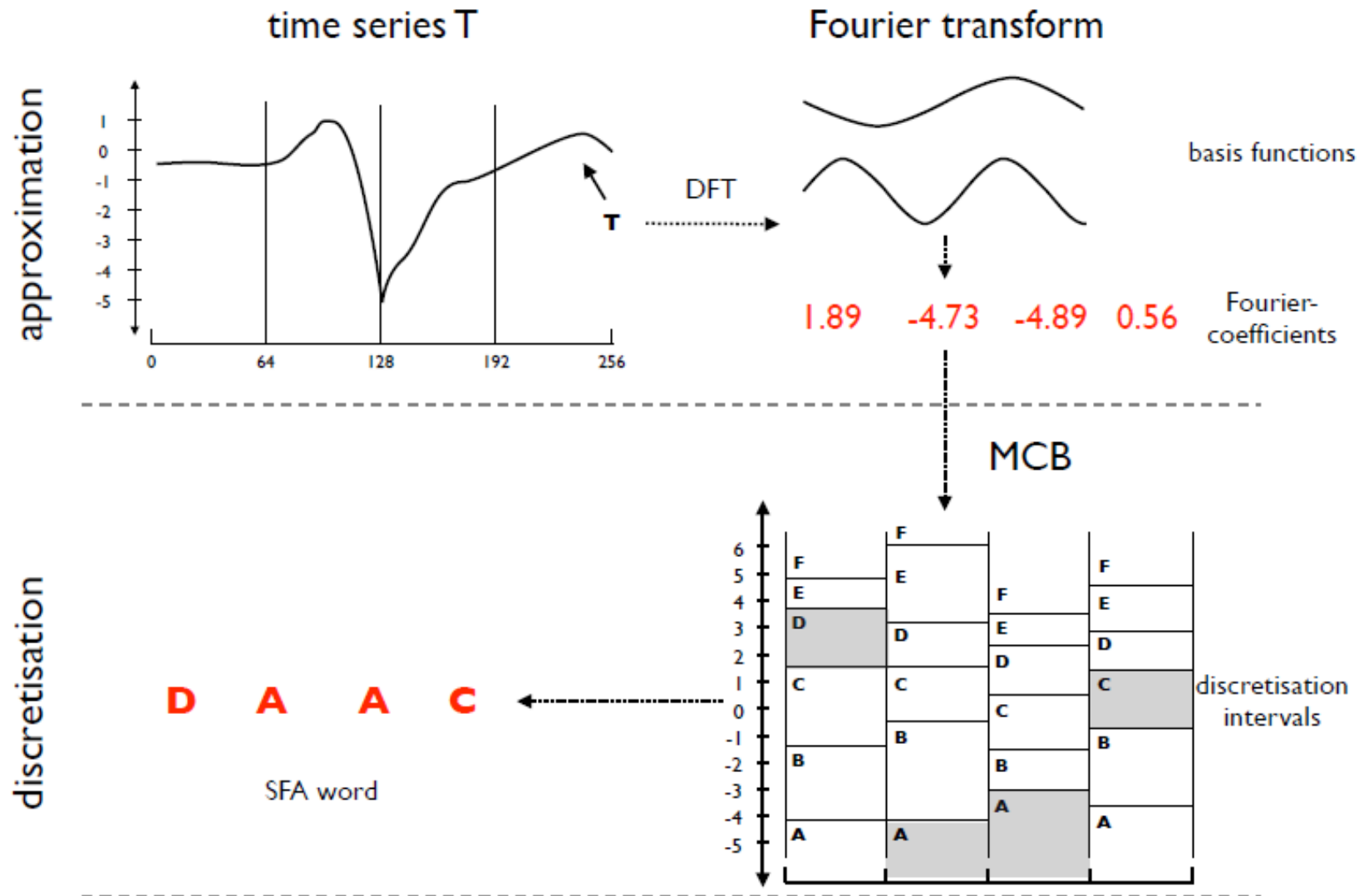
While shapelets look for the presence of a pattern, dictionary methods look for pattern repetition.



SAX Transform



SFA Transform



Domain agnostic classification methods: Co-eye^[1] and Red CoMETs^[2]

- A multi-resolution ensemble classifier for Symbolically Approximated Time Series
- Looking at time series from different perspectives similar to flies' compound eye.
- Each eye consists of many ommatidia, each one is an individual eye by itself.



Abdallah, Z.S. and Gaber, M.M. (2020). Co-eye: a multi-resolution ensemble classifier for symbolically approximated time series. *Machine Learning*, 109(11), pp.2029-2060.

Bennett, L.A., Abdallah, Z.S. (2023). RED CoMETs: An Ensemble Classifier for Symbolically Represented Multivariate Time Series. In: Ifrim, G., *et al.* *Advanced Analytics and Machine Learning on Temporal Data. AALTD 2023. Lecture Notes in Computer Science()*, vol 14343.

Lens: A symbolic representation with either Fourier or piecewise approximation, using specific parameters for zooming in and out

$S_i \gg \text{Symbolic}(w_i, \alpha_i)$

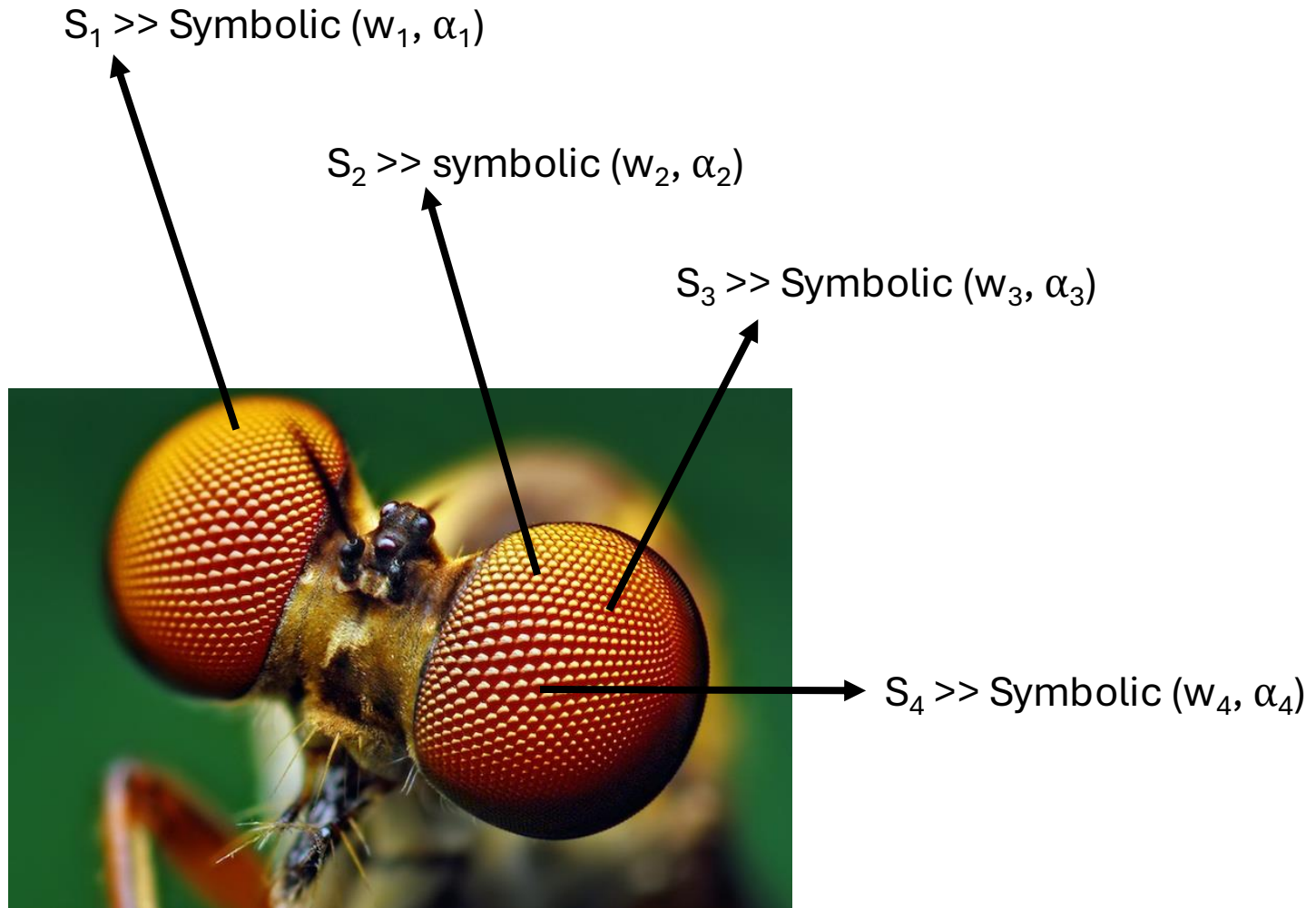


Eye: A classifier trained using a lens that has specific values for the triplet

$C_i \gg \text{Classifier}(S_i)$



A compound Eye: A collection of classifiers of different lenses

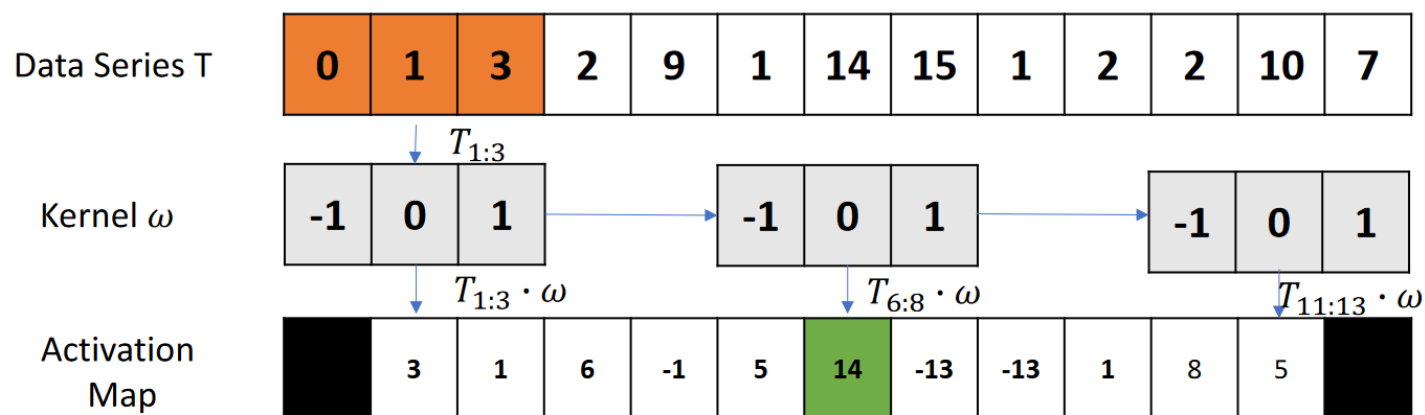
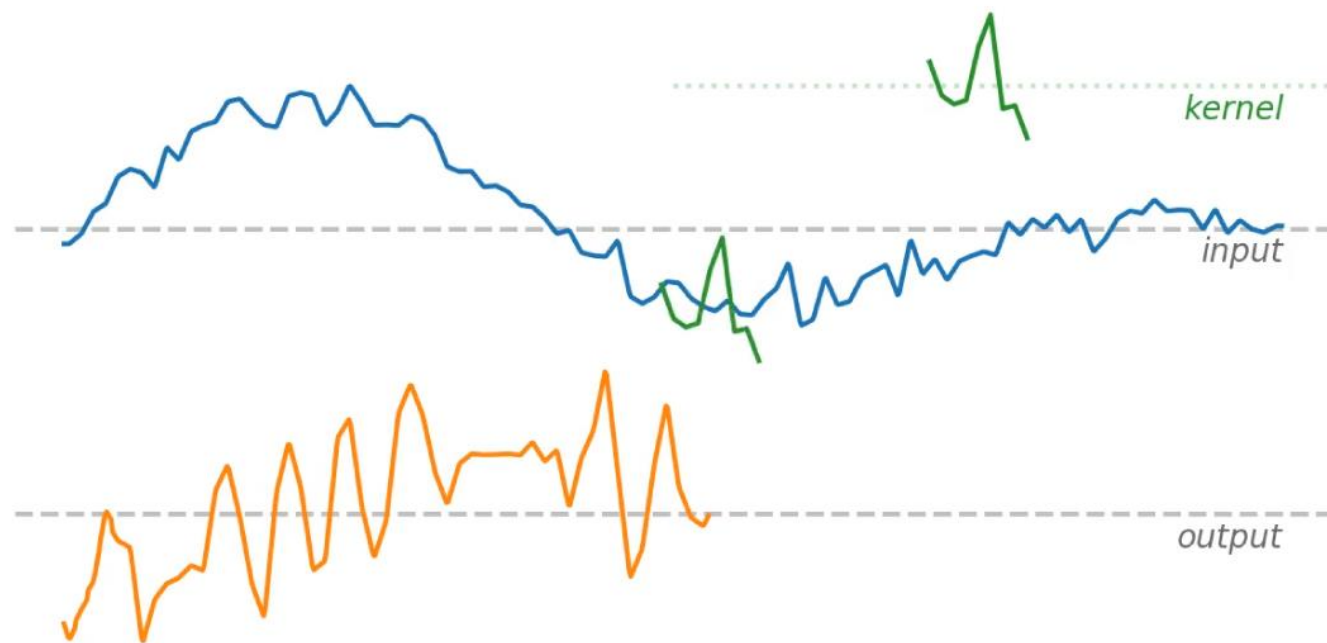


Convolution-based Algorithms

Convolution approaches used convolutional kernels with pooling operations to generate feature vectors.

These kernels are usually random, not learnt. Not deep learning!

Related to shapelets running a sliding window over the full series and pooling. Key difference is that shapelets use subseries extracted from the training data and distances.



Convolution-based Algorithms

Features are extracted from the kernel output series using **pooling operations** i.e. the ROCKET classifier uses:

Max

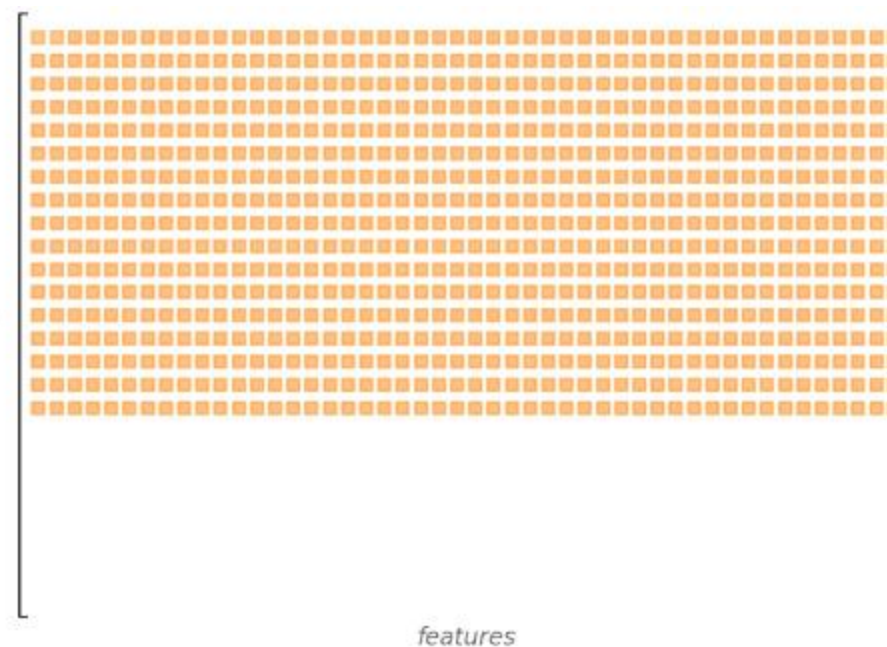
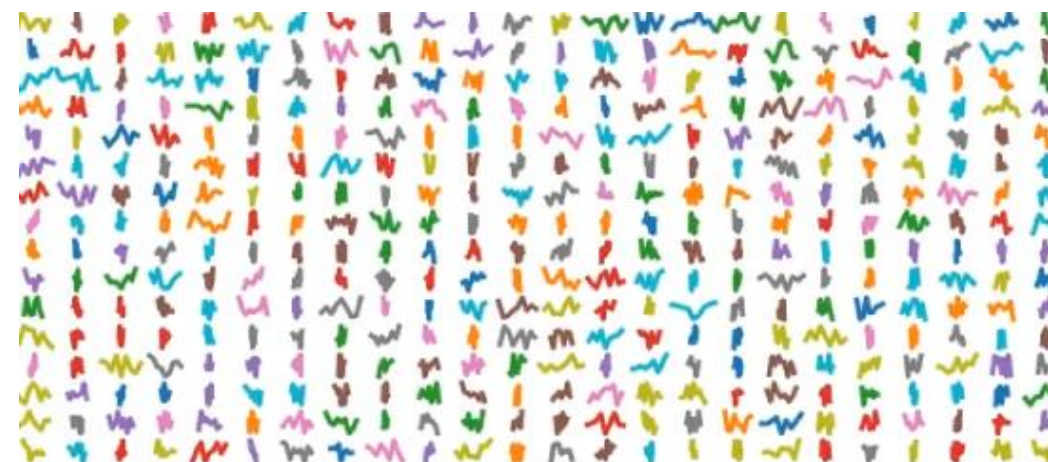
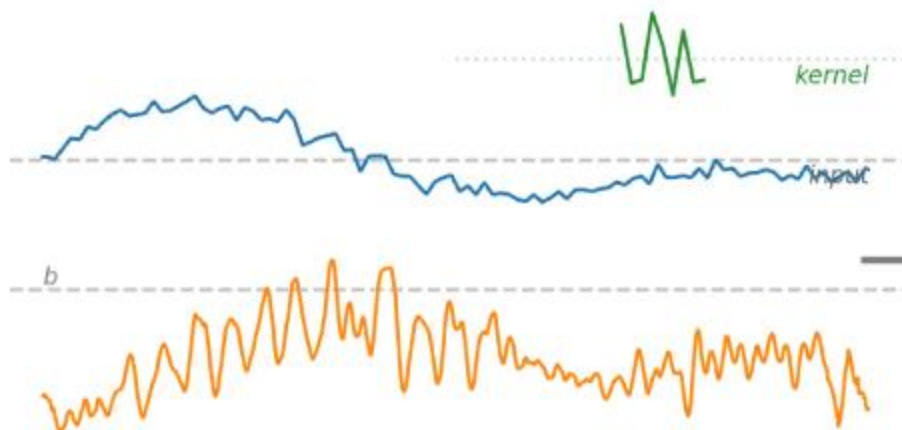


Proportion of Positive Values (PPV)



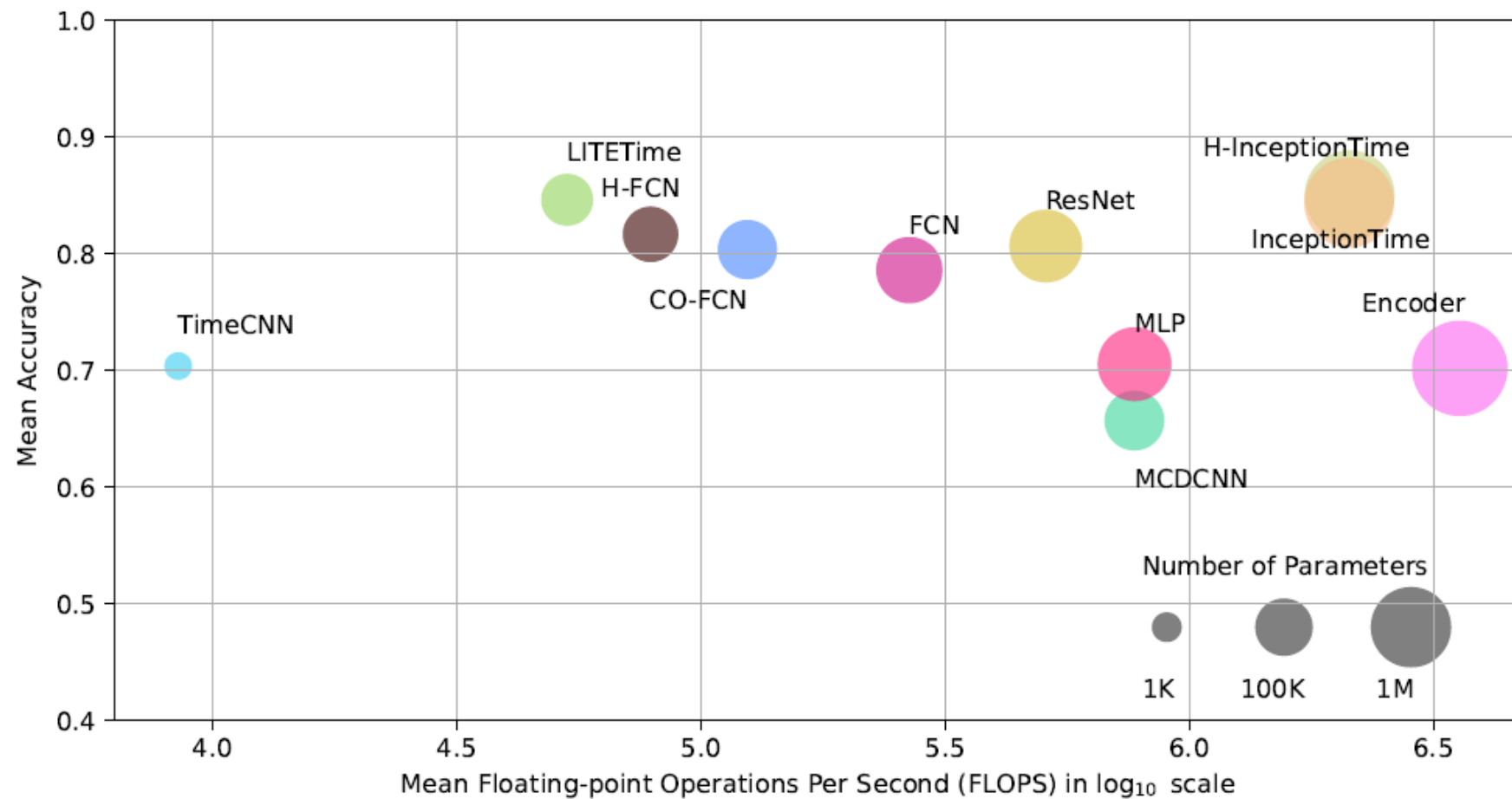
ROCKET

The ROCKET classifier applies thousands of randomly generated kernels (length, weights, bias, dilation, and padding) and builds a linear classifier with the pooled output.

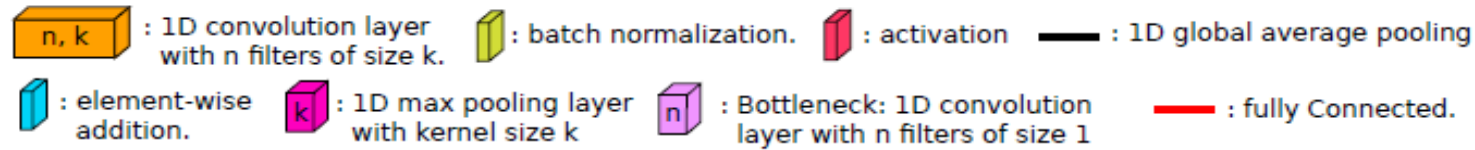


linear classifier

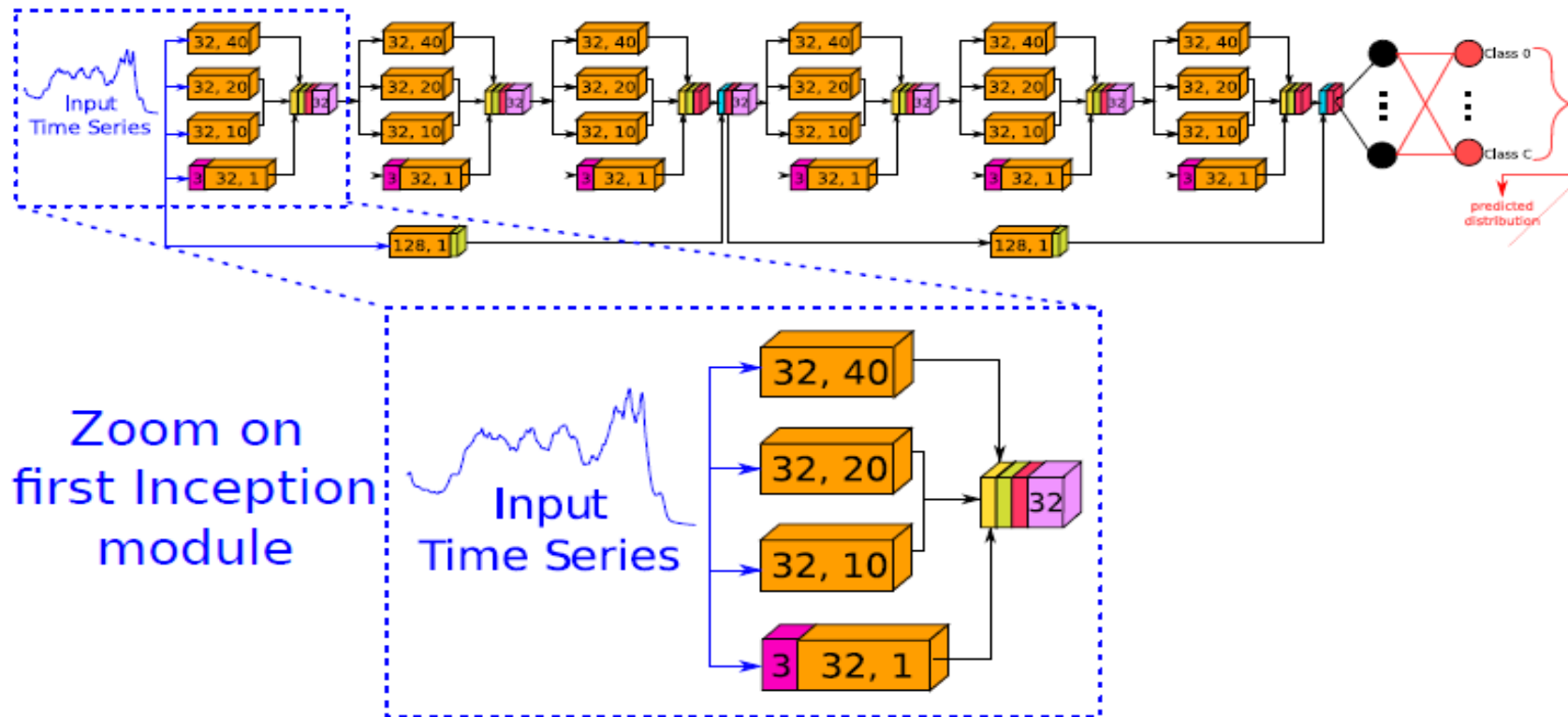
Deep Learning for TSC/TSER



Inception Time

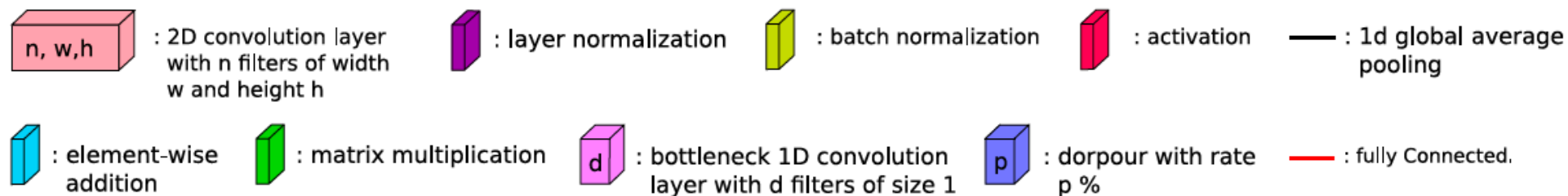


Inception architecture for TSC

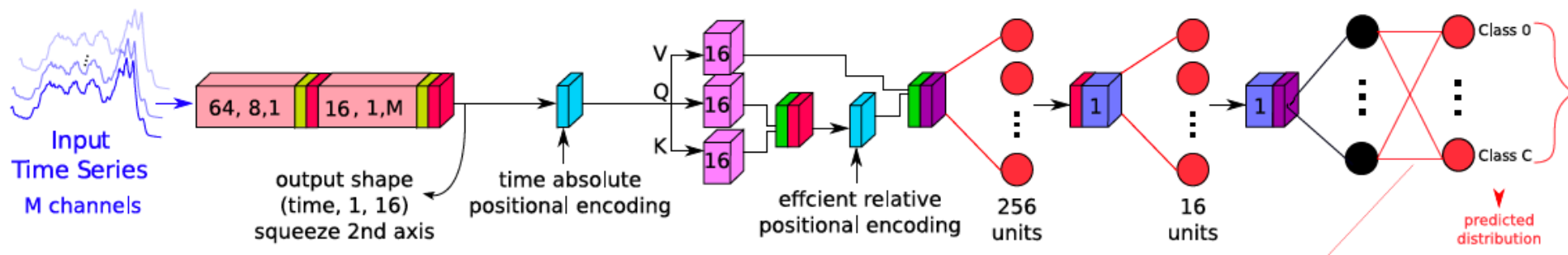


Deep learning for multivariate time series

Architecture using Transformer:



ConvTran Architecture



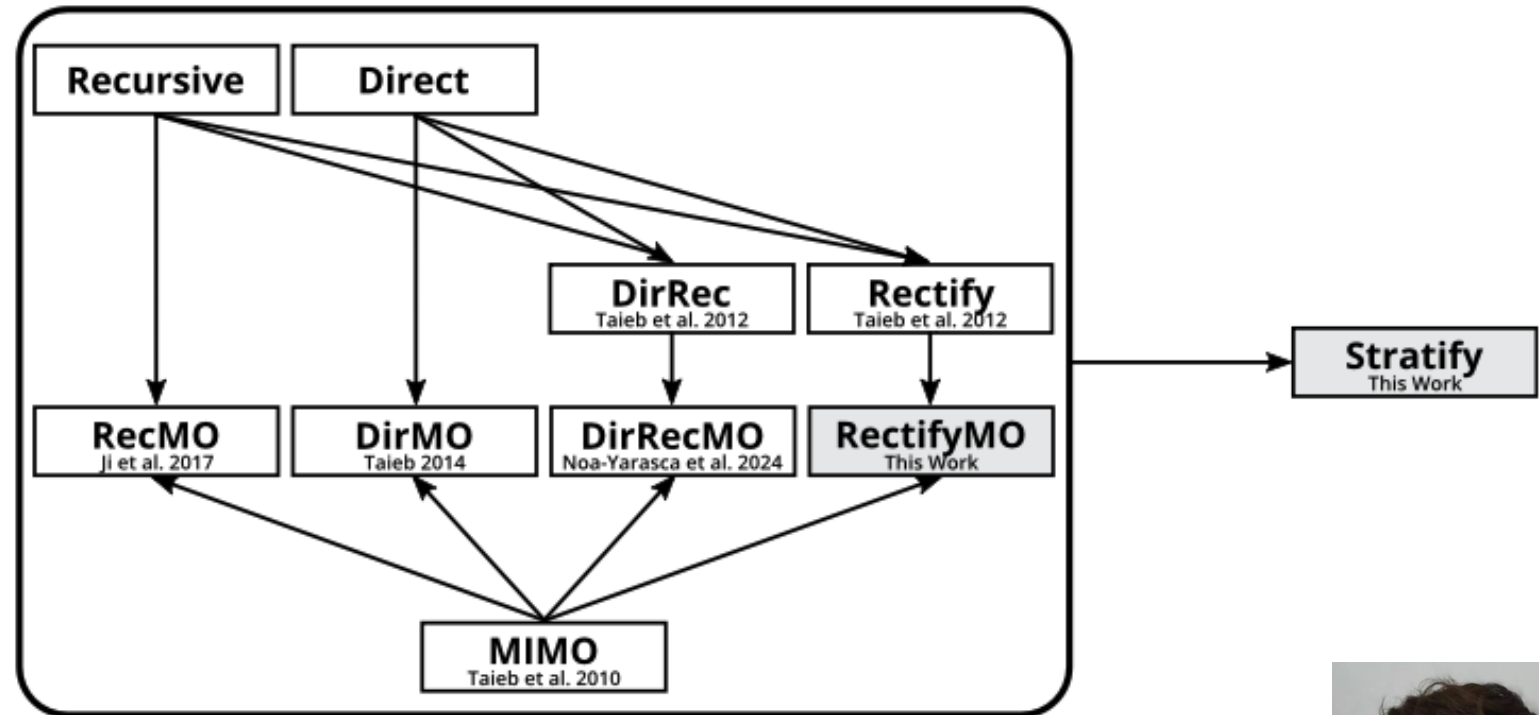
 Foumani, Navid Mohammadi, et al. "Improving position encoding of transformers for multivariate time series classification." *Data Mining and Knowledge Discovery* 38.1 (2024): 22-48.

New directions in TSC/ TSER

- Representation learning
- Foundation models for TS
- Specialised models for specific type of time series (such as EEG)
- TSC and active learning
- TSC with concept drift
- Open world TS
- LLM for TSC

Time series (main) tasks

Stratify: unifying multi-step forecasting strategies



Green, Riku, Grant Stevens, Zahraa S. Abdallah, and Telmo M. Silva Filho. "Stratify: unifying multi-step forecasting strategies." *Data Mining and Knowledge Discovery* 39, no. 5 (2025): 1-39.

Green, R., Day, H., & Abdallah, Z. S. and Telmo M. Silva Filho (2025). Epistemic Error Decomposition for Multi-step Time Series Forecasting: Rethinking Bias-Variance in Recursive and Direct Strategies. *arXiv preprint arXiv:2511.11461*.

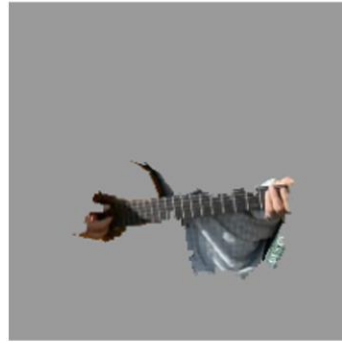


Explainability in Images

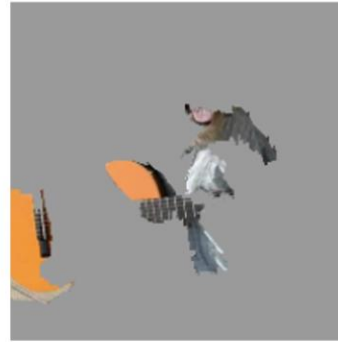
Lime explanation of super pixel (Ribeiro et. al, 2016)



(a) Original Image



(b) Explaining *Electric guitar*



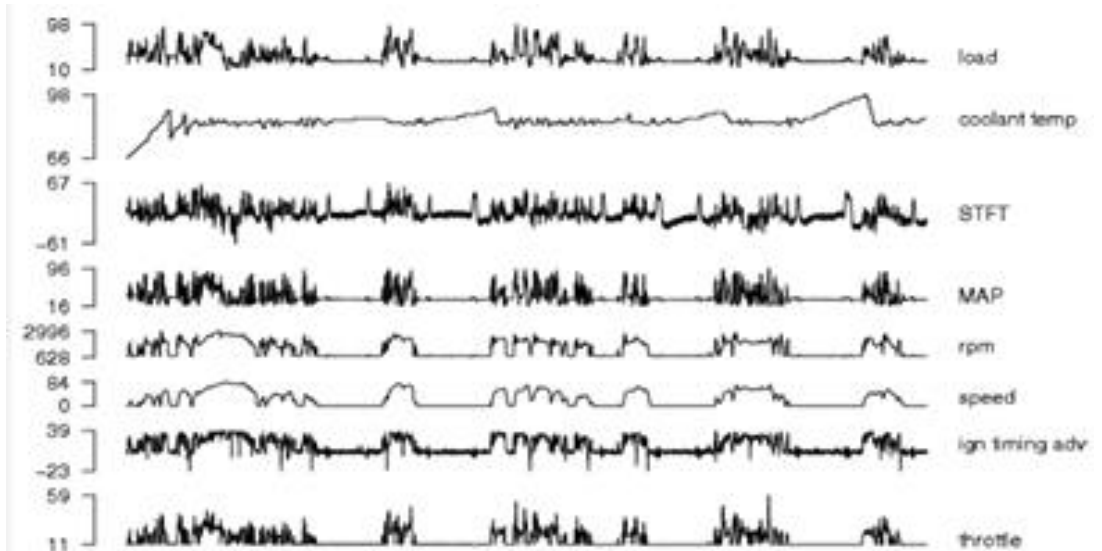
(c) Explaining *Acoustic guitar*



(d) Explaining *Labrador*

Explainability in Time Series

A faulty signal, why?



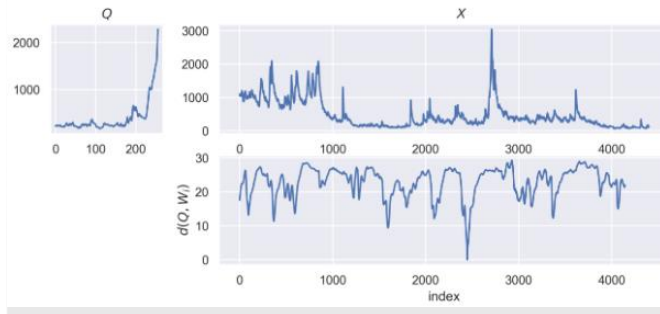
recording of vehicle signals, taken from (Theissler, 2013)

Time series (main) tasks



https://github.com/davide-serramazza/segment_SHAP

Explainability



SHAP

LIME

Saliency
Maps

Counterfactual
Analysis

Temporal
Decomposition

An Empirical Evaluation of Factors Affecting SHAP Explanation of Time Series Classification

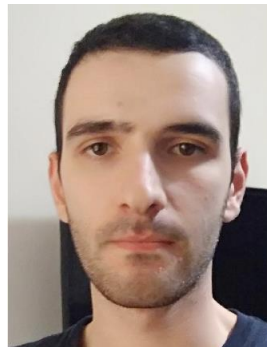
Nikos Papadeas^{1*}, Davide Italo Serramazza (✉)^{2*}, Zahraa Abdallah¹, and Georgiana Ifrim²

¹ School of Engineering Mathematics and Technology, University of Bristol, UK

² School of Computer Science, University College Dublin, Ireland

*These authors contributed equally to this work.

Abstract. Explainable AI (XAI) has become an increasingly important topic for understanding and attributing the predictions made by complex Time Series Classification (TSC) models. Among attribution methods, SHapley Additive exPlanations (SHAP) is widely regarded as an excellent attribution method, but its computational complexity, which scales exponentially with the number of features, limits its practicality for long time series. To address this, recent studies have shown that aggregating features via segmentation, to compute a single attribution value for a group of consecutive time points, drastically reduces SHAP running time. However, the choice of the optimal segmentation strategy remains an open question. In this work, we investigated eight different Time Series Segmentation algorithms to understand how segment compositions

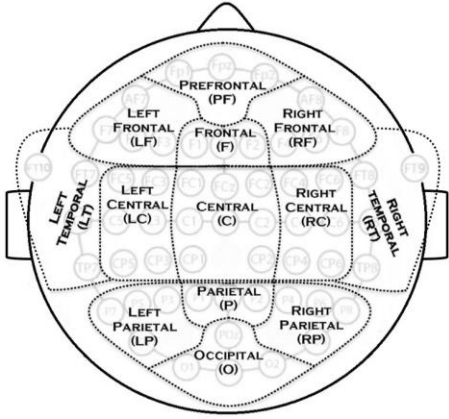
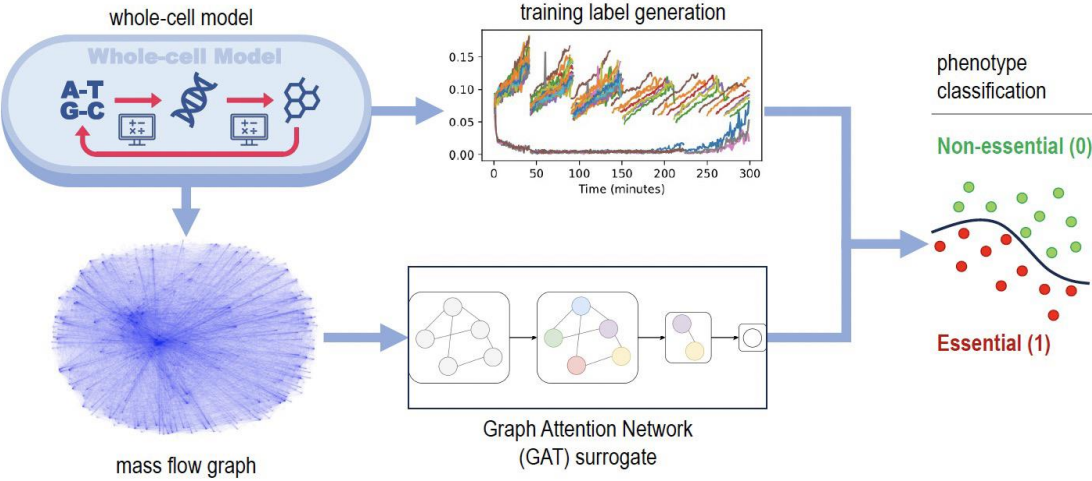


Time Series for healthcare

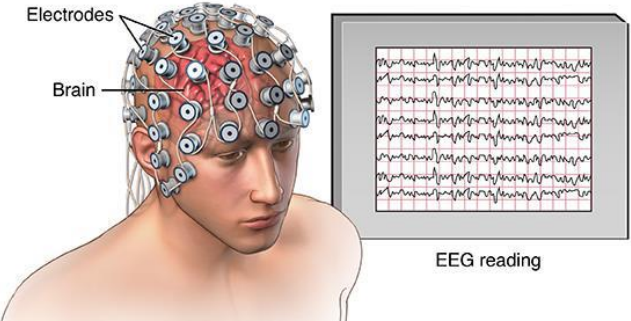


Classification of Early-Stage Parkinson's Disease from EEG

Timeseries for Gene Essentiality



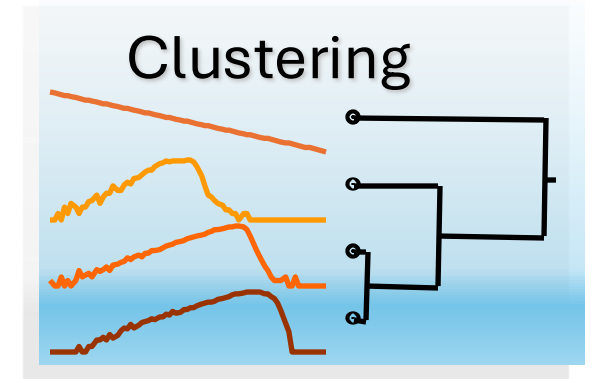
Electroencephalogram (EEG)



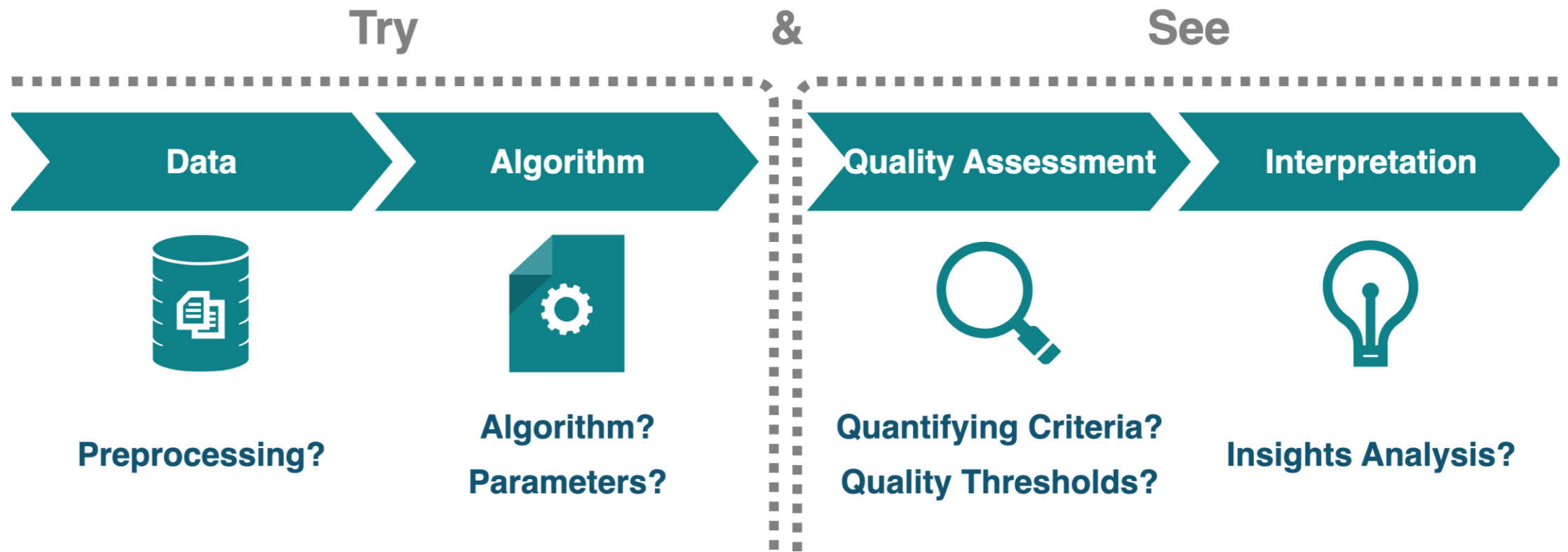
Sharma, K., Marucci, L., & Abdallah, Z. S. (2024). Fluxgat: Integrating flux sampling with graph neural networks for unbiased gene essentiality classification. arXiv preprint arXiv:2403.18666.

Sahota, Amarpal, Santos-Rodriguez, R., Abdallah Z.S. (2024), "A Time Series Approach to Parkinson's Disease Classification from EEG." , AAAI-24 Health intelligence workshop
Sahota, Amarpal, Navid Mohammadi Foumani, Raul Santos-Rodriguez, and Zahraa S. Abdallah. "KnowEEG: Explainable Knowledge Driven EEG Classification." arXiv preprint arXiv:2505.00541 (2025).

Time Series Clustering



- Different methods are developed for time series clustering:
 - Shape based methods & Feature based methods & Model based methods & Representation Learning & DL Methods
- Benchmarking studies: 8 algorithms with 3 distance metrics (Javed et al., 2020); 80 univariate methods on UCR datasets (Holder et al., 2024)
- The main insight is **performance variation depended more on dataset characteristics than algorithmic innovations.**



“Clustering represents an expansive domain in which practitioners must develop customised approaches for each application context” (Jaeger & Banks, 2023)

Clustering: Art or Science?

(von Luxburg et al., 2012)

Rethinking Clustering

- Is it about the **distance metric**, **choosing k** , or something else?
- If “ground truth” = class labels, are we **evaluating clustering** or just **approximate classification**?
- Do algorithms **reveal structure** or merely **optimise a criterion**?
- How valid are **spherical/convex shape** assumptions?
- How **trustworthy** are our validation indices?
- What role should **domain expertise** play across the pipeline?

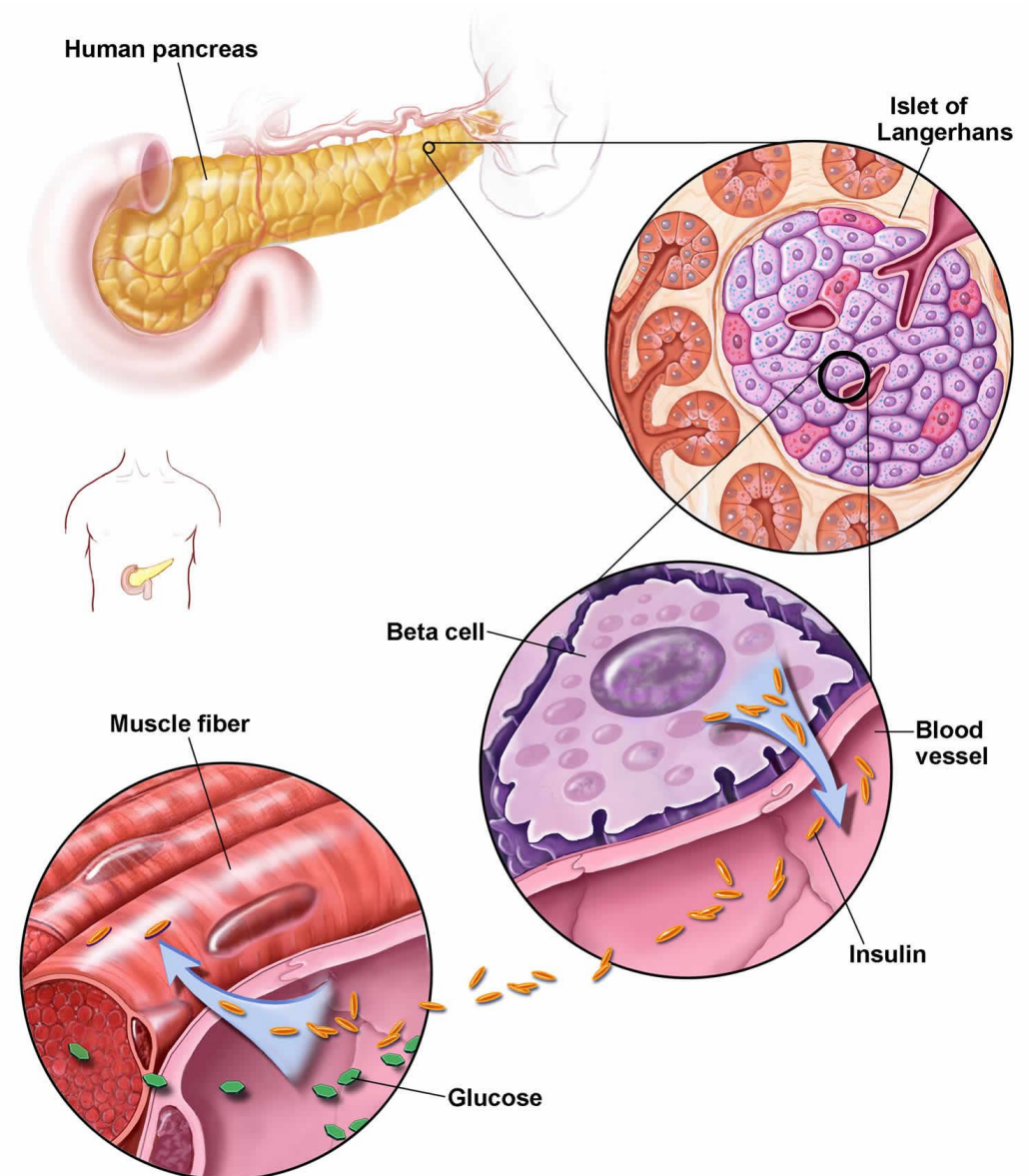


Journey from Pattern Discovery to Validation Paradox

Use Case- Type1 Diabetes

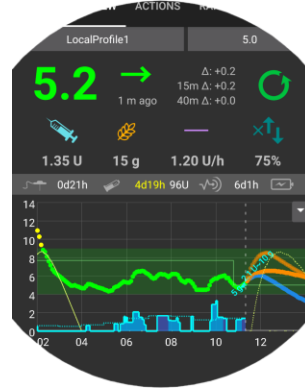
Type 1 Diabetes (T1D)

- The pancreas produces too little or no insulin, the hormone that regulates blood glucose.
- **Treatment:** Exogenous (injected) insulin.
- **Goal:** Keep blood glucose **stable and within range** to avoid complications. **How?**



Type I Diabetes Technology

Continuous
Glucose
Monitor



Closed-loop
algorithm



Insulin
Pump

How much and when to inject insulin?

Regulating insulin remains a complex control task that often
goes wrong

BrisT1D Blood Glucose Prediction Competition

Using historical blood glucose readings, insulin dosage, carbohydrate intake, and smartwatch activity data to predict future blood glucose.

[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#)

Overview

Predicting blood glucose fluctuations is crucial for this can alleviate some of the challenges faced

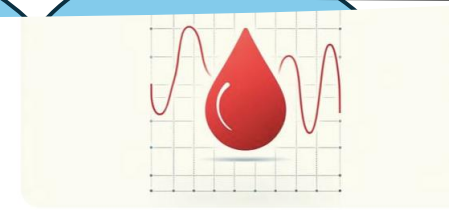
Goal: Forecast blood glucose levels one hour ahead

Start

Sep 18, 2024



Sam Gordon James



going to
hour?

I going to eat



BrisT1D-Open Dataset

The BrisT1D Dataset features the data collected in a longitudinal study in which 24 young adults with Type 1 Diabetes (T1D) in the UK were given smartwatches to use alongside their T1D self-management. During the six-month study, participants donated the data collected by their T1D devices and smartwatches and were involved in monthly interviews or focus groups. The anonymised transcripts of these study sessions are included along with the device data, in both an anonymised raw state and a processed state.

This is the Open-Access part of the BrisT1D Dataset. It includes the processed state device data, study forms, transcripts, and demographic data. The raw state device data can be found in the BrisT1D-Restricted Dataset (<https://doi.org/10.5523/bris.yonrplcb4bvi2vhhn2ehtmpcy>), available through data.bris (University of Bristol Data Repository) under Restricted Access.

[Complete download \(zip, 150.3 MiB\)](#)

CSTS: A Benchmark for the Discovery of Correlation Structures in Time Series Clustering

Isabella Degen¹, Zahraa S Abdallah², Henry W J Reeve³, and Kate Robson Brown⁴

¹School of Computer Science, University of Bristol

²School of Engineering Mathematics and Technology, University of Bristol

³School of Artificial Intelligence, University of Nanjing

⁴College of Engineering and Architecture, University College Dublin

Position: Clustering Results Are Unfalsifiable Without Structure-Specific Validation

Establishing Validity for Distance Functions and Internal Clustering Validity Indices in Correlation Space

Isabella Degen¹, Zahraa S Abdallah², Kate Robson Brown³, and Henry W J Reeve⁴



Beyond Expected Patterns in Insulin Needs of People With Type 1 Diabetes: Temporal Analysis of Automated Insulin Delivery Data

Isabella Degen¹ ; Kate Robson Brown² ; Henry W J Reeve³ ; Zahraa S Abdallah⁴

Temporal patterns in insulin needs for Type 1 diabetes

Isabella Degen

Department of Computer Science
University of Bristol
Bristol, BS8 1TH

isabella.degen@bristol.ac.uk

Zahraa S. Abdallah

Department of Engineering Mathematics
University of Bristol
Bristol, BS8 1TH

zahraa.abdallah@bristol.ac.uk

OpenAPS Data Commons

Expected Patterns

High IOB → IG should **fall** (or flatten)

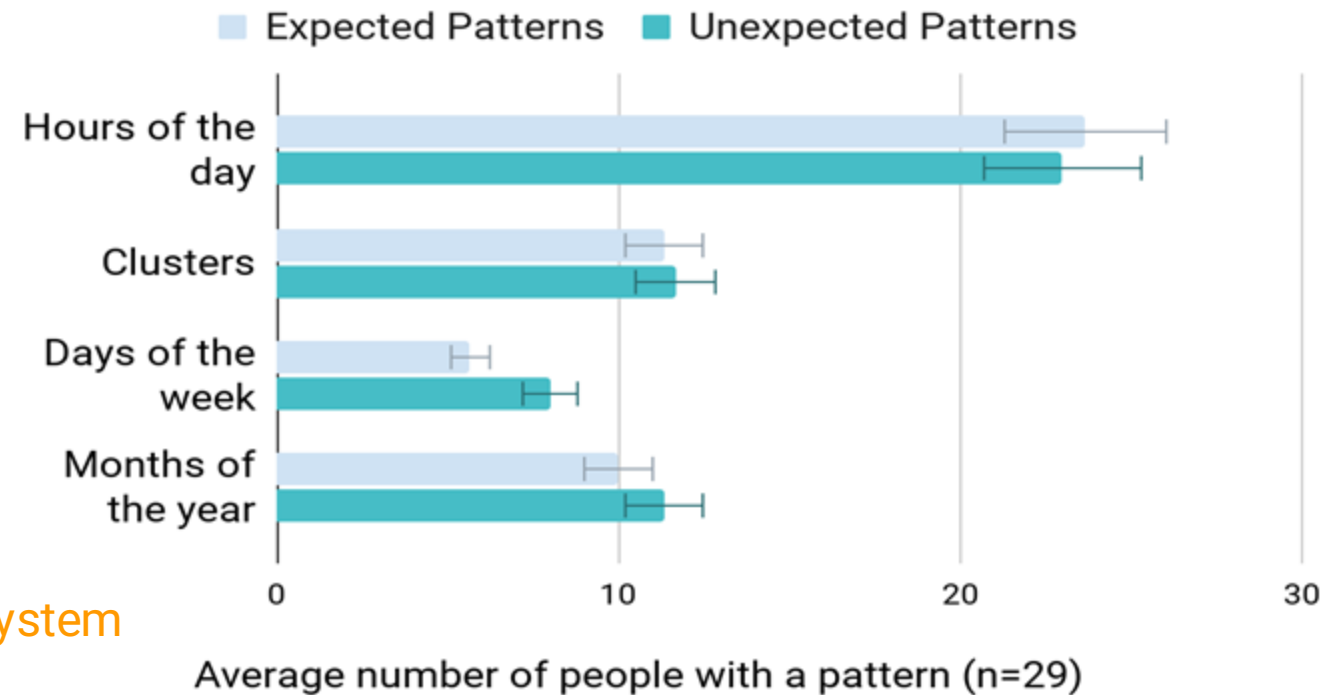
High COB → IG should **rise**

- open-source automatic insulin delivery data
 - System build by individuals with T1D
- Collected in free-living conditions & donated to research

IOB → Insulin on board, calculated by the system

COB → Carbohydrates on board, calculated by the system

IG → interstitial glucose, from the sensor



Unexpected patterns in insulin needs are as frequent as expected ones

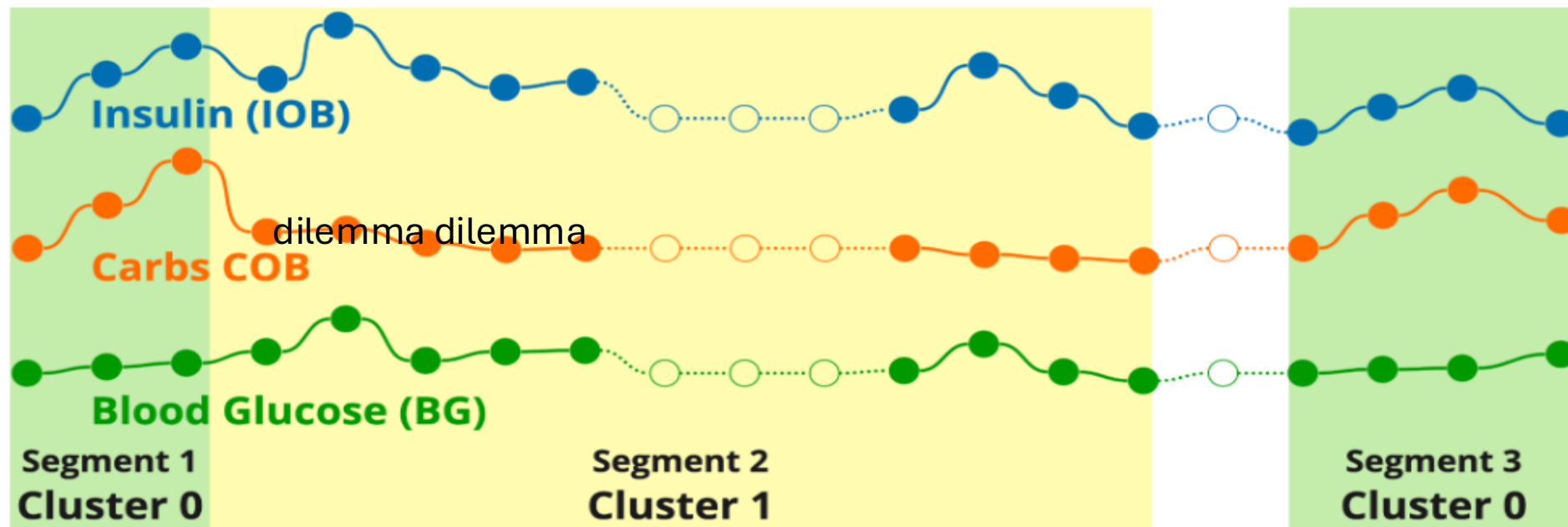
Revealing New Patterns

Unexpected Patterns

... increasing insulin is not lowering blood glucose

... eating carbohydrates is not rising blood glucose

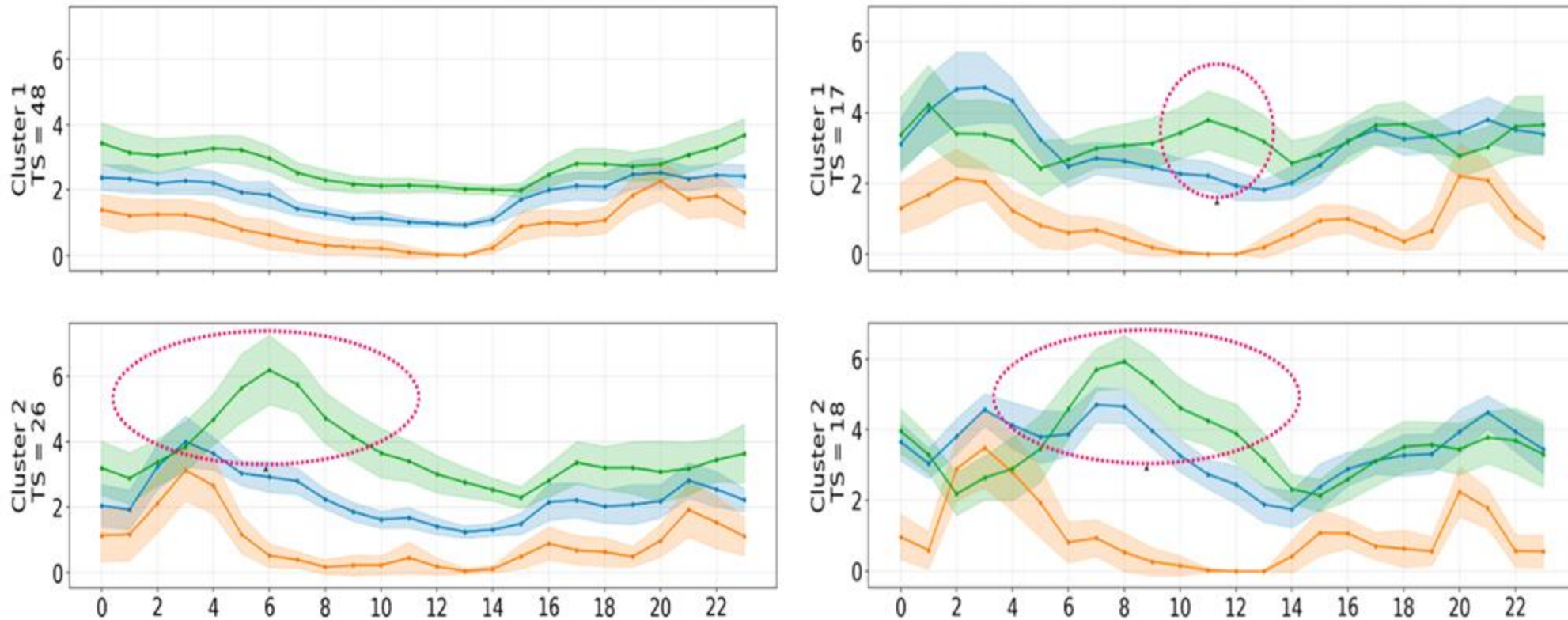
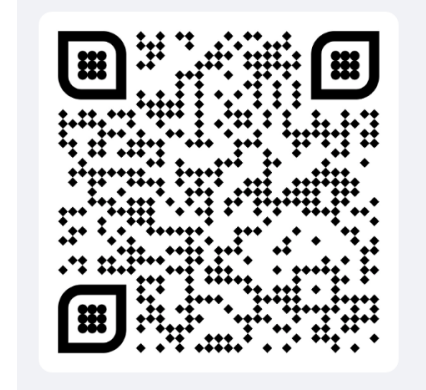
Pattern Discovery in Type 1 Diabetes



Unexpected patterns are a sign that underexplored factors are at play

- Degen, I., Brown, K. R., & Abdallah, Z. S. (2023). Studying insulin needs in Type 1 Diabetes by analysing the OpenAPS Data Commons. *International Journal of Population Data Science*, 8(3)
- Degen, Isabella, Kate Robson Brown, Henry WJ Reeve, and Zahraa S. Abdallah. "Beyond expected patterns in insulin needs of people with type 1 diabetes: temporal analysis of automated insulin delivery data." *Jmirx med* 5 (2024): e44384.

High IG at night despite significantly higher IOB



Significantly higher IOB didn't avoid significantly higher IG

Domain Guided Discovery

- Limit new discoveries
- Expensive
- Highly reliant on manual post hoc interpretation and domain knowledge

Alternative?



Apply suitable clustering method for correlated time series that can cluster and segment at the same time.

Key Approaches

- K-mean/ k-medoids with different distance measures.
 - Most popular Euclidean distance (ED) or Dynamic Time Warping (DTW)
 - Holder et al. (2024): best results with **Move–Split–Merge (MSM)** and **Time Warp Edit (TWE)** distances
- Toeplitz Inverse Covariance-Based Clustering (Hallac et al., 2017)
 - Multivariate time series clustering
 - Segmentation and clustering
 - Correlated time series

Toeplitz Inverse Covariance-Based Clustering (TICC)

$$\operatorname{argmin}_{\Theta \in \mathcal{T}, \mathbf{P}} \sum_{i=1}^K \left[\overbrace{\|\lambda \circ \Theta_i\|_1}^{\text{sparsity}} + \sum_{X_t \in P_i} \left(\overbrace{-\ell\ell(X_t, \Theta_i)}^{\text{log likelihood}} + \overbrace{\beta \mathbb{1}\{X_{t-1} \notin P_i\}}^{\text{temporal consistency}} \right) \right]$$

Θ : k Toeplitz inverse covariance matrices
 $\mathbf{P}$: cluster assignment for each observation

Hyper-parameters:

number of clusters $\mathbf{k}$, window size $\mathbf{n}$,
sparsity of Θ via λ and penalise cluster
switches via β

Algorithm: Initialise Θ and $\mathbf{P}$

Iterate until $\mathbf{P}$ is stationary then return Θ and $\mathbf{P}$:

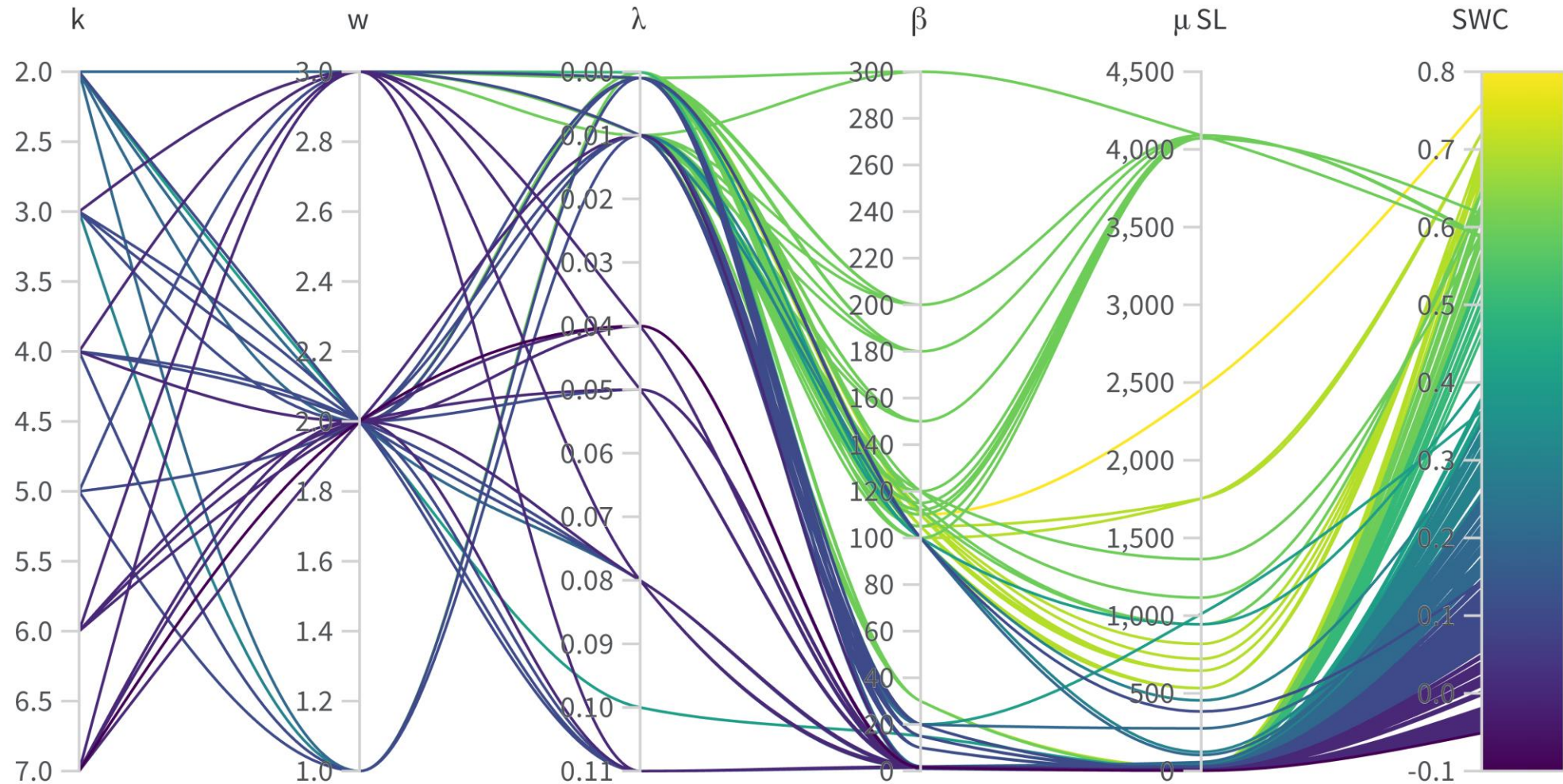
E-step: update $\mathbf{P}$, keep Θ fixed (viterbi path)

X_t are observations at time t

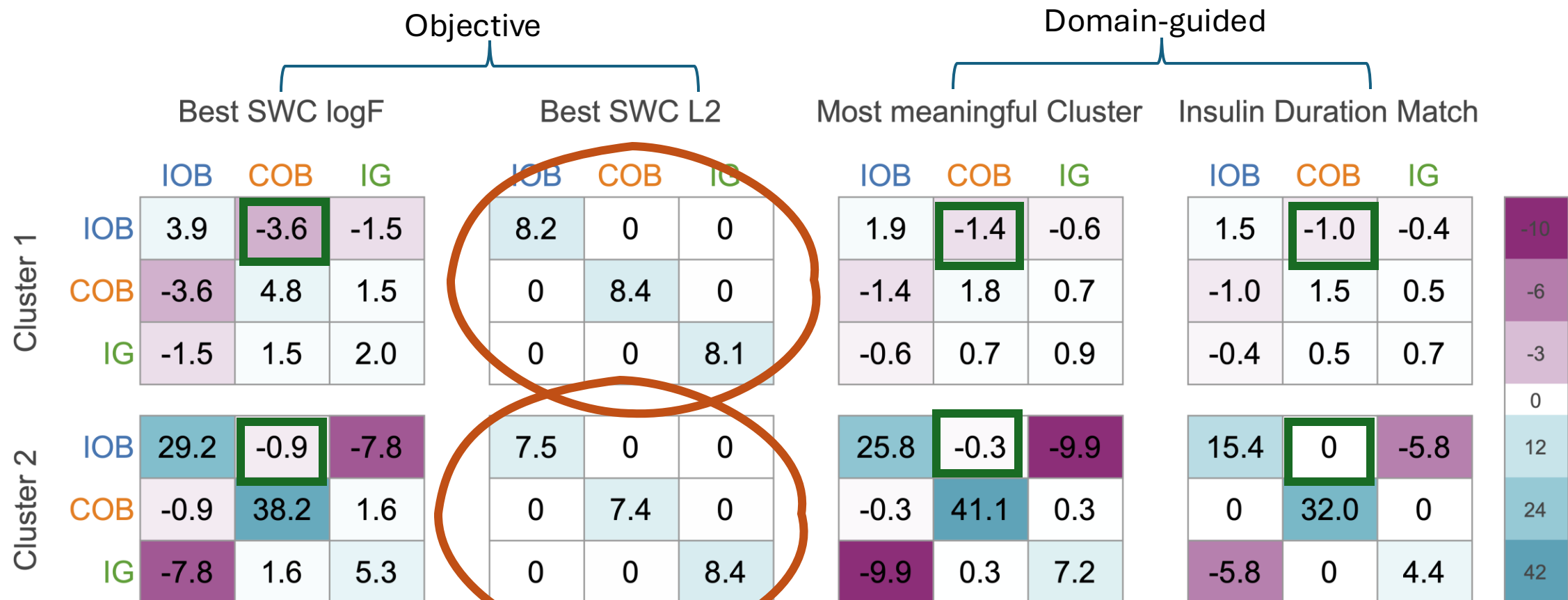
M-step: update Θ , keep $\mathbf{P}$ fixed (ADMM solver)

i is cluster, S are empirical covariances

Parameters Selection for TICC



Existing methods, success?



Genuine insights or algorithm artefacts?

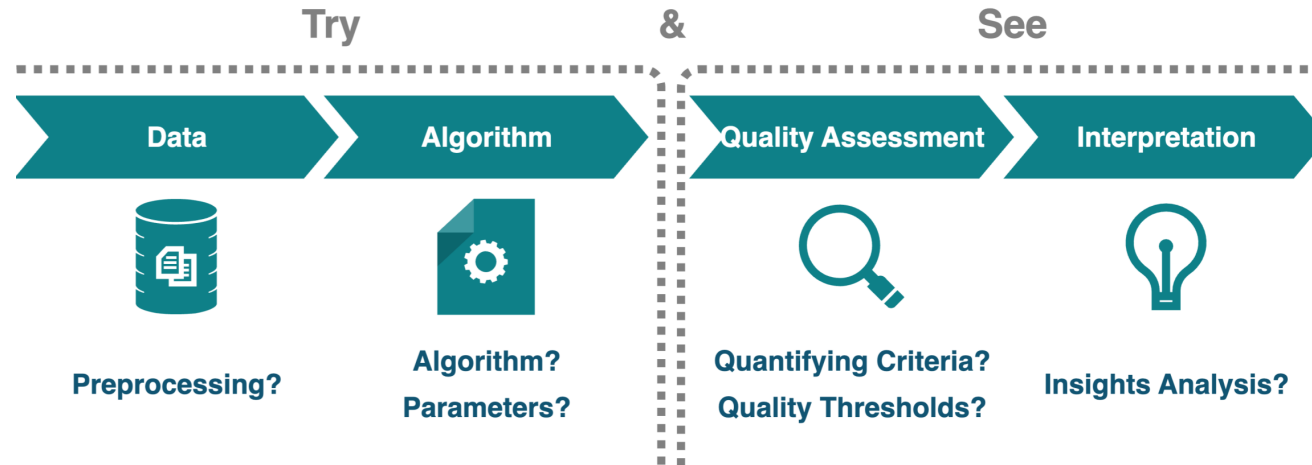
Clustering Validation Paradox!

We were unable to validate which clustering outcomes were superior and which were inferior.

whether our results were showing meaningful patterns or merely methodological artefacts and how we could tell the difference.

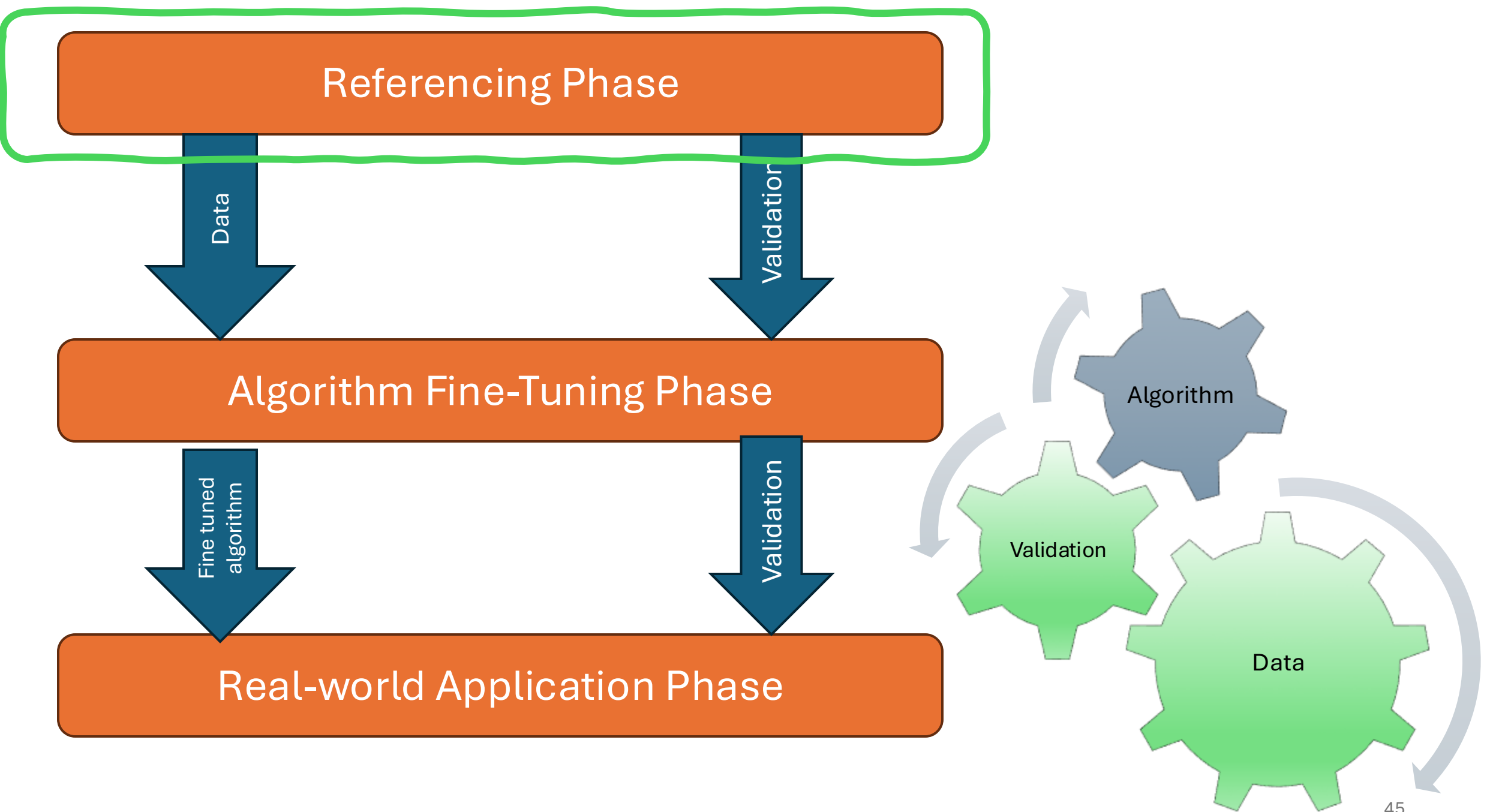


Clustering as an Exploratory Tool



More objective questions

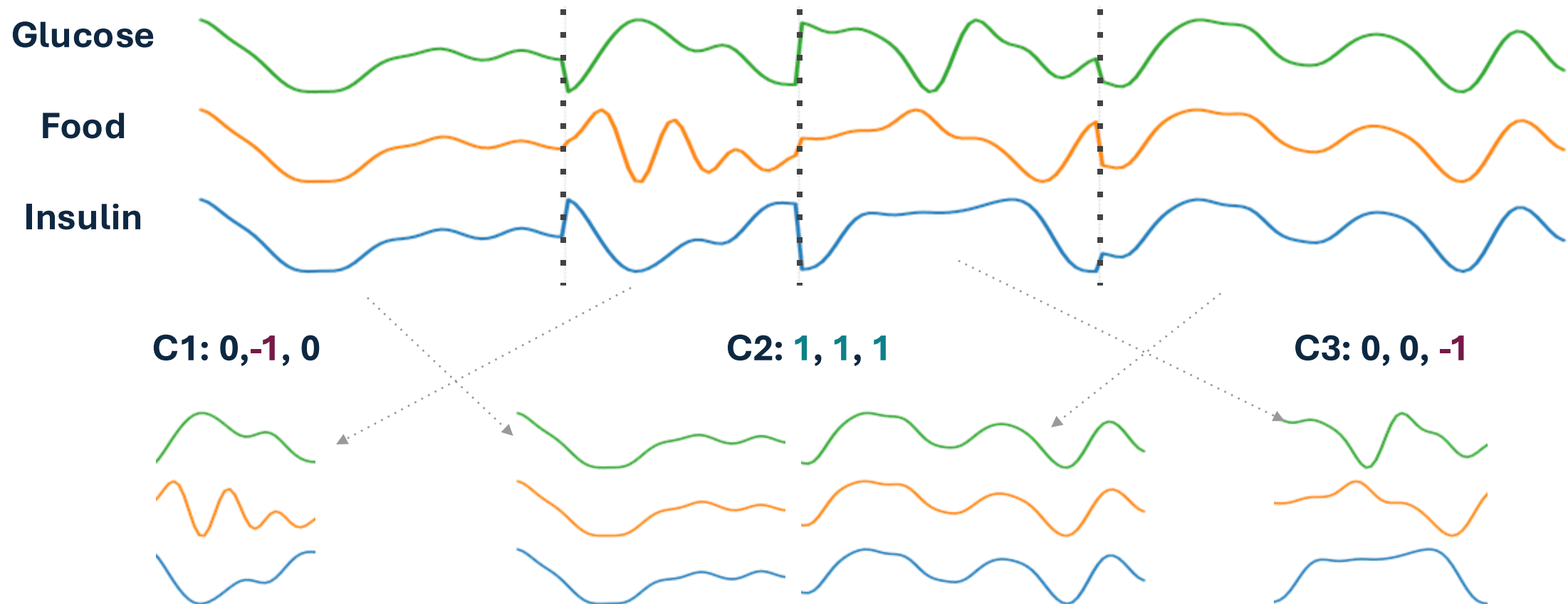
- What is the **structure** of the data?
- What **indices** would be considered good for chosen structure?
- Which **distance** function should we use?
- Are these patterns **meaningful**?



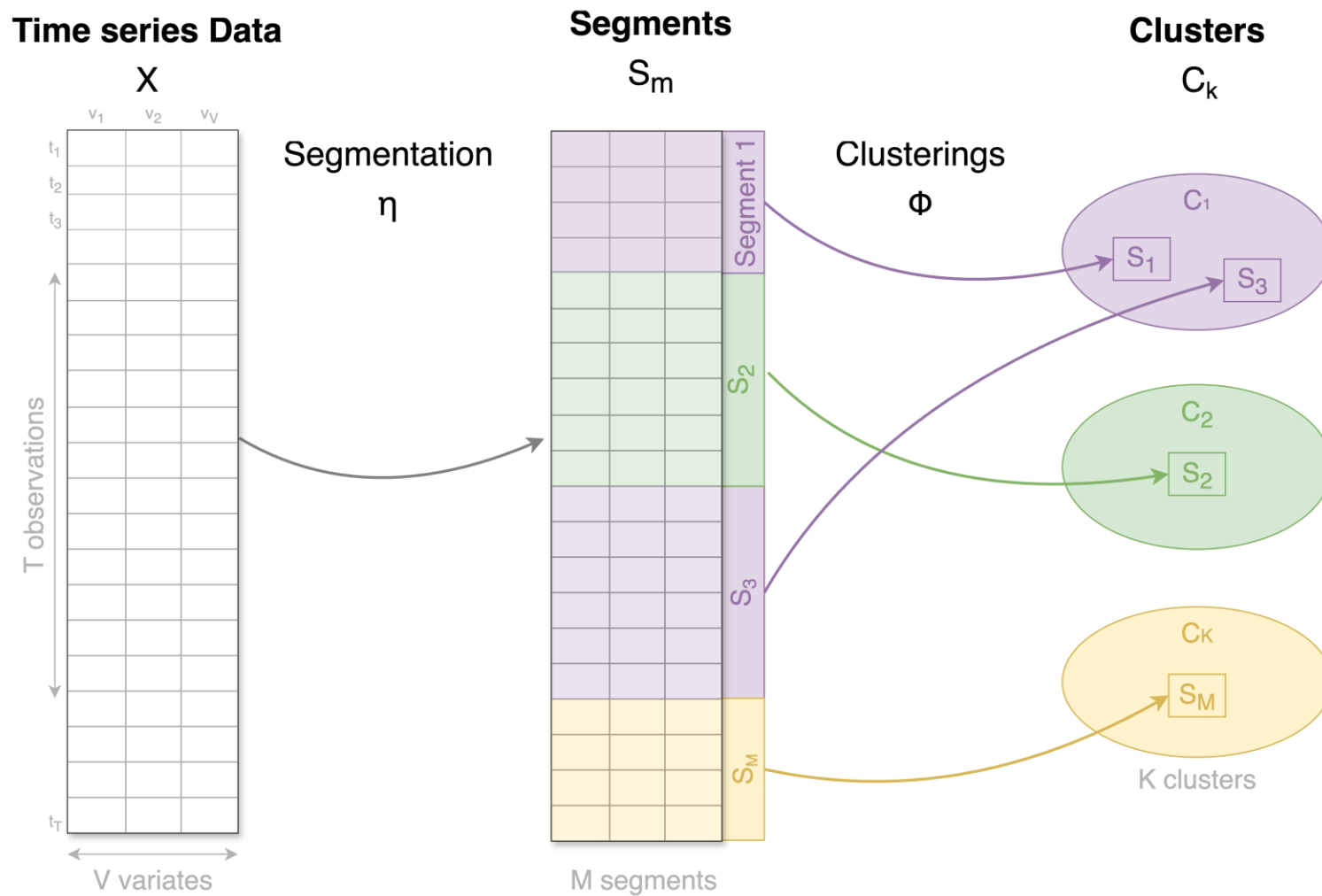
A Seemingly Easy Problem

Data: multivariate time series Type 1 Diabetes treatment data (3 variables)

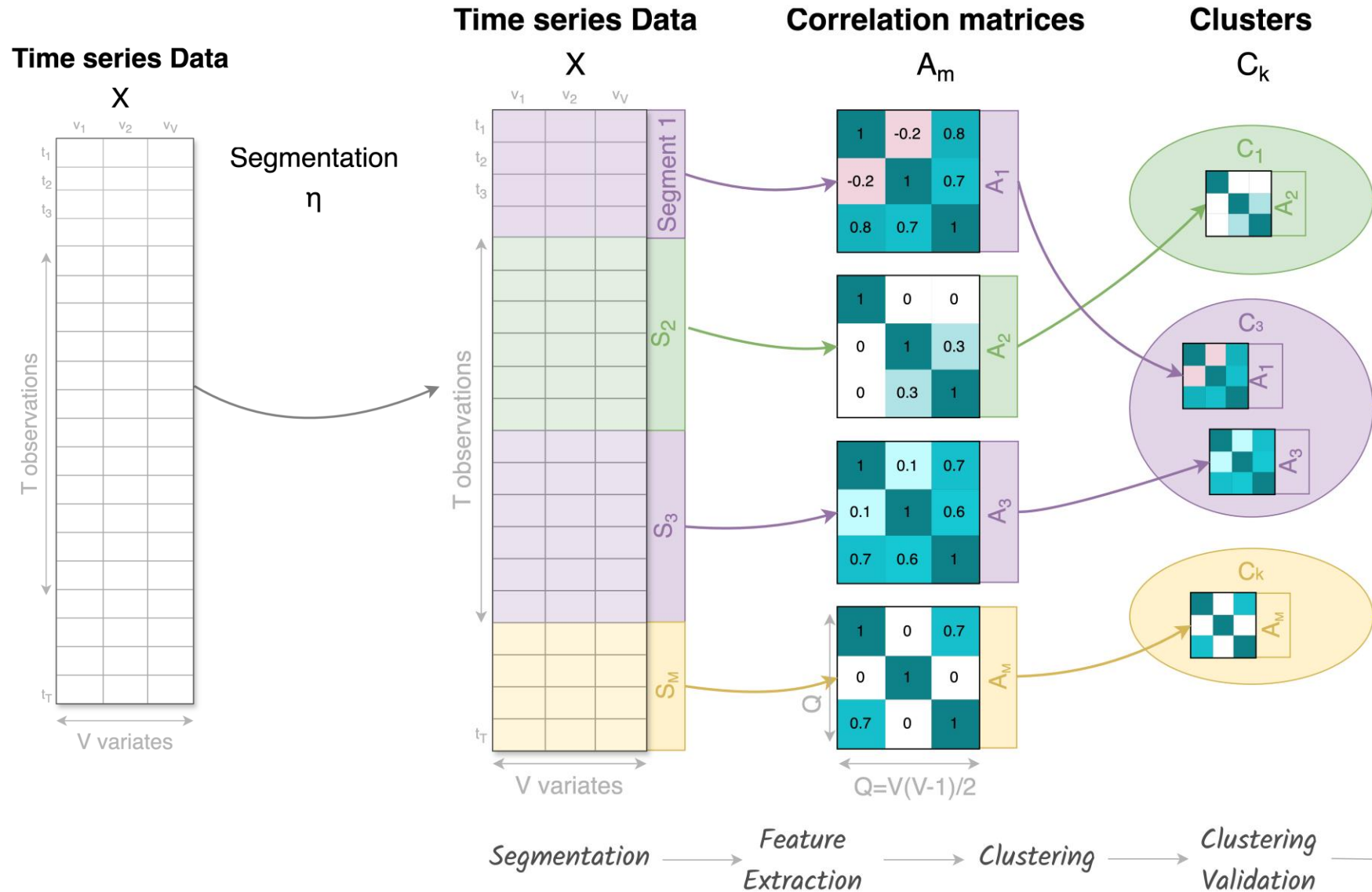
Task: segment and group into different correlation regimes (ρ_{12} , ρ_{13} , ρ_{23})



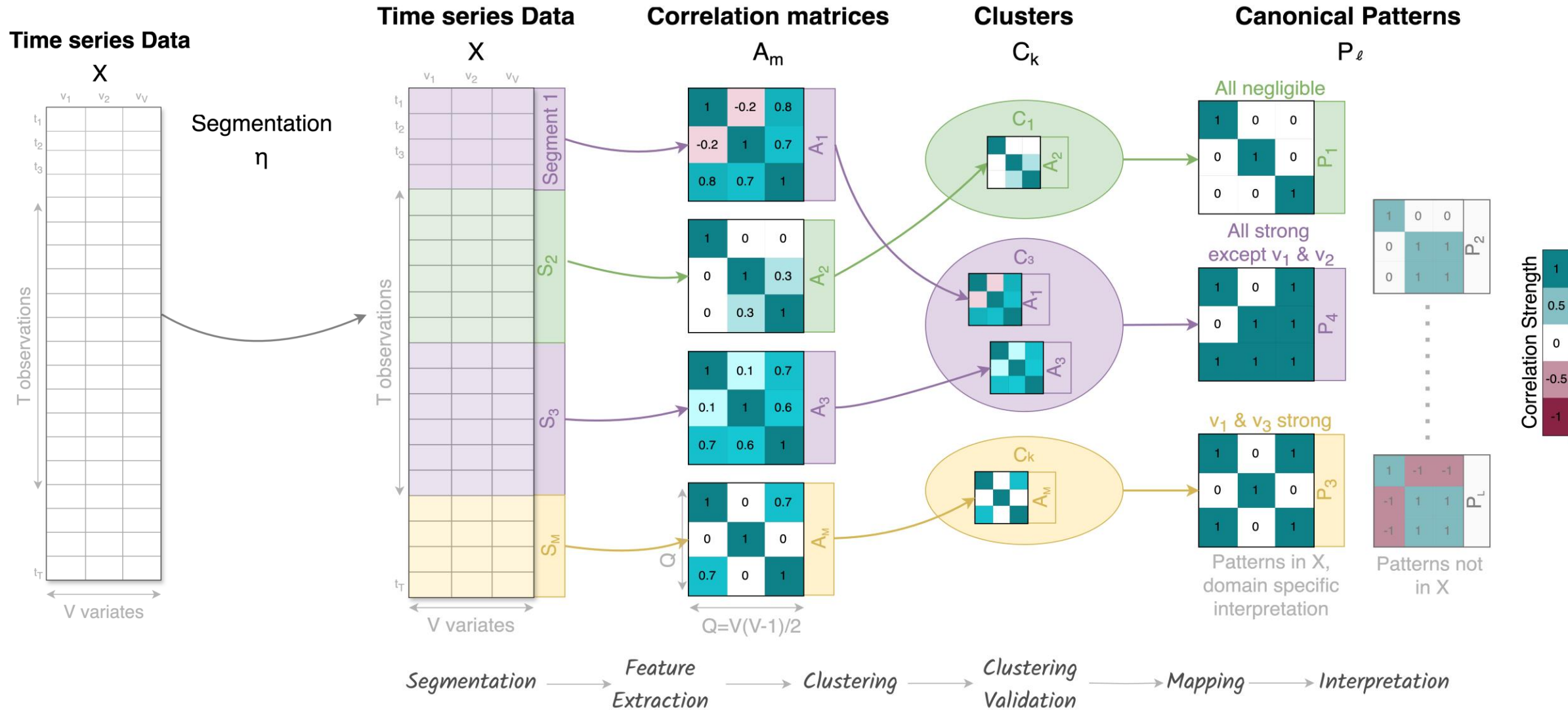
Time series clustering — and segmentation



Correlation-based Clustering

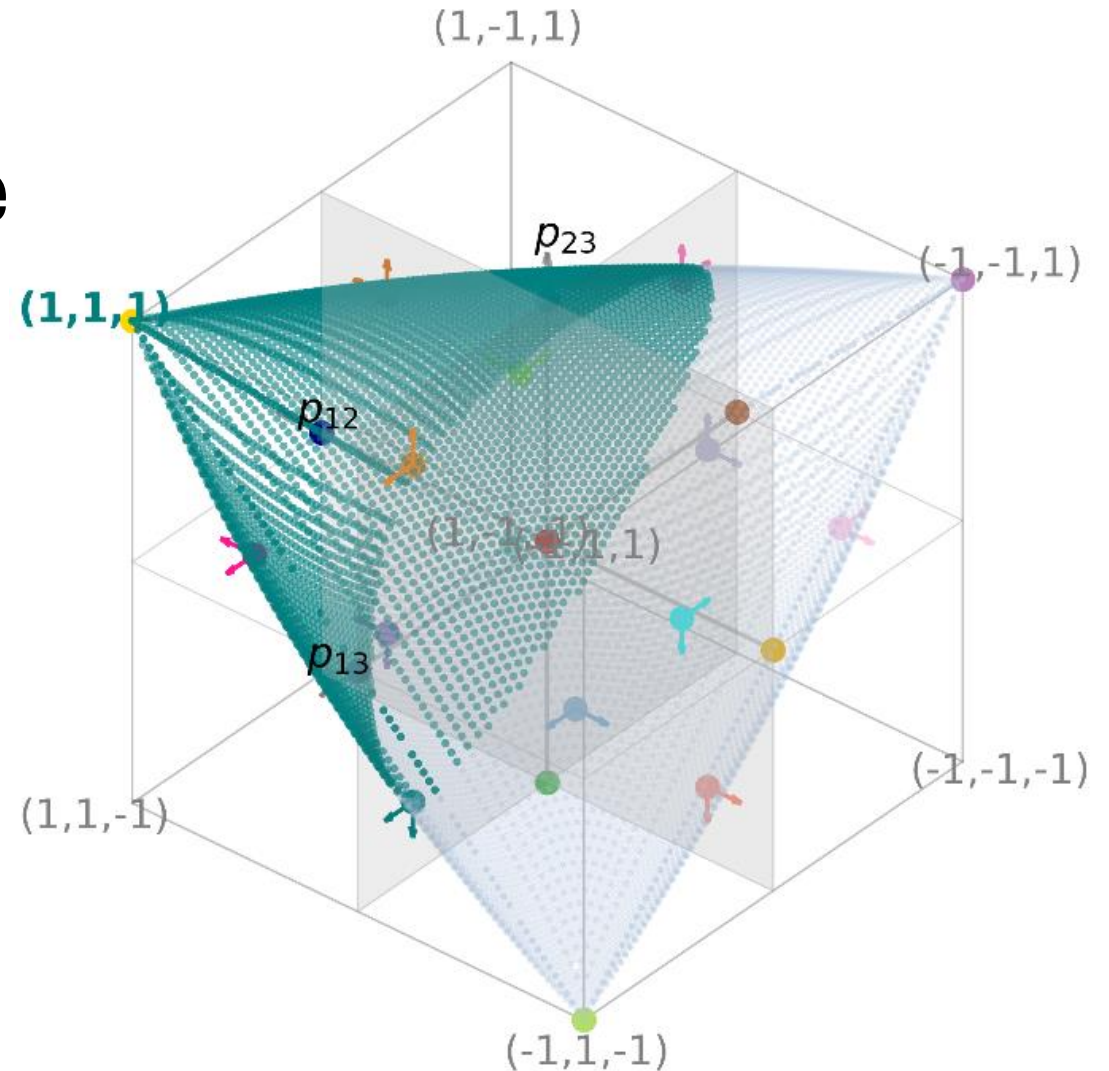


Correlation-based Clustering



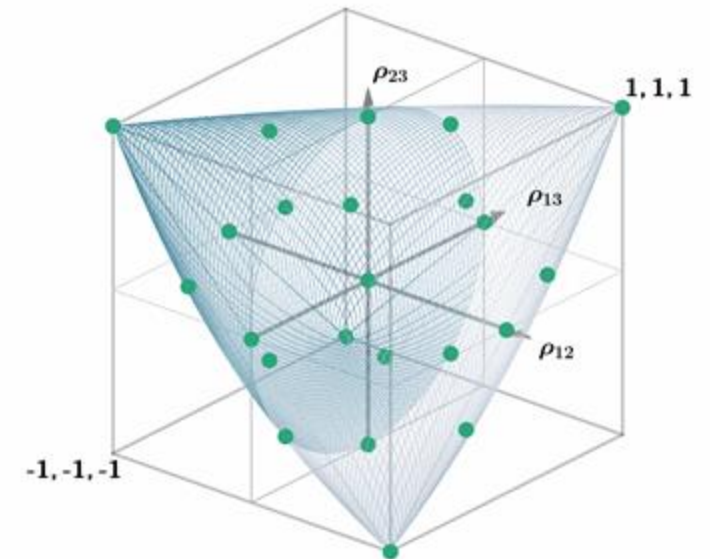
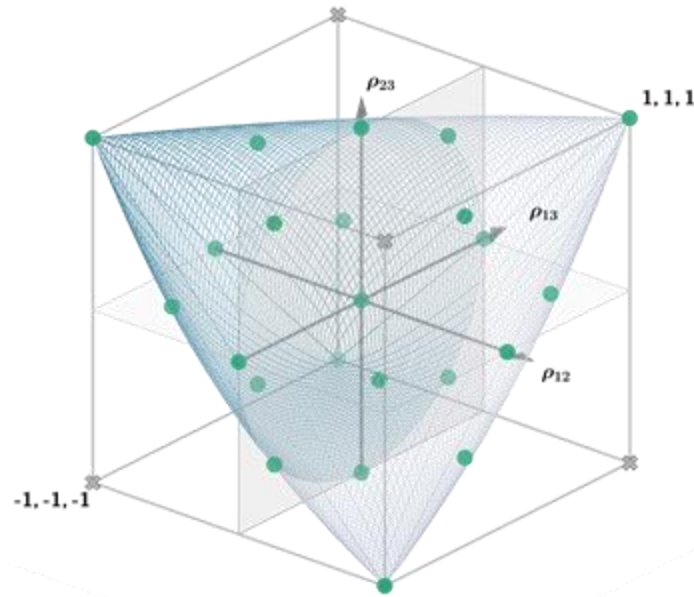
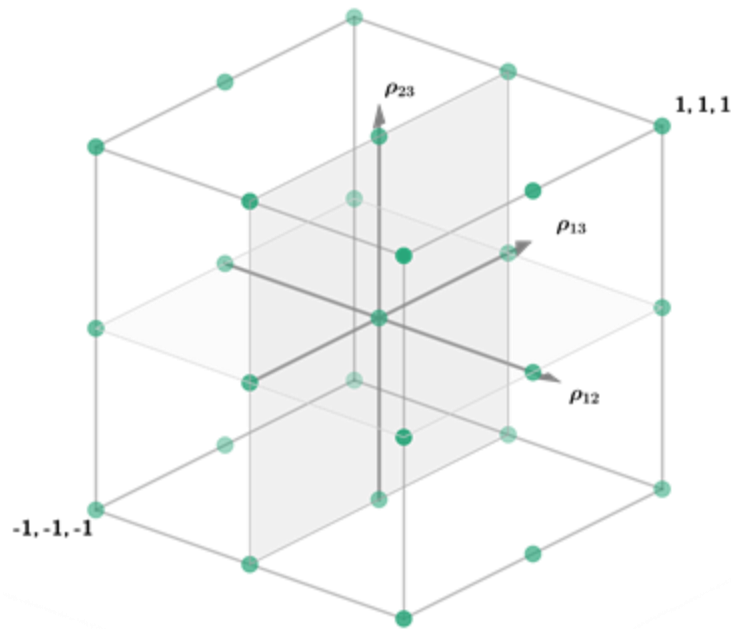
Break the Circle by Defining Optimal Structure

- A canonical pattern P_ℓ discretise the infinite correlation space into mathematically defined states
- states **Elliptope** E_n is the set of all **valid** canonical patterns P_ℓ .
- Model all combinations of strong positive (1), negligible (0), and strong negative (-1) correlation coefficients



Elliptope, a curved geometric space defined by the positive semi-definite (PSD) constraint, with smooth manifold structure in the interior but singularities at the boundary

Break the Circle by Defining Optimal Structure



All 27 combinations of $\{1, -1, 0\}$ PSD constraint $\rightarrow$ 23 valid **canonical correlation patterns**

CSTS (**C**orrelation **S**tructures in **T**ime **S**eries)

- 60 subjects (30 exploratory and 30 confirmatory)
- Each subject consists of 100 segments of various length varies from 15 mins-10 hours (900-36000 observation at 1HZ sampling).
- Each canonical pattern is used 4-5 times per subject with varying segment length.
- Each pattern across all subjects >> minimum of 240 segments of varied length.
- 12 variation per subject, Includes controlled data variations: distribution shifts, sparsification, downsampling, degraded clustering quality.
- Total of 1.26M observation per subject

Datasets: [idegen/csts](#)  like 0

Modalities: [Tabular](#) [Text](#) [Time-series](#) Formats: [parquet](#) Size: [100M - 1B](#) ArXiv: [arxiv:2505.14596](#)

Tags: [timeseries](#) [timeseries clustering](#) [changepoint-detection](#) [correlation-structure](#) [Synthetic](#) [benchmark](#) Libraries: [Datasets](#) [Dask](#) [Croissar](#)

License: [cc-by-4.0](#)

[Dataset card](#) [Data Studio](#) [Files and versions](#) [xet](#) [Community](#) 1



Dataset Viewer [Auto-converted to Parquet](#) [API](#) [Embed](#) [Data Studio](#)

Subset (36)
nonnormal_complete_data · 188M rows

Split (10)
exploratory · 37.9M rows

subject_id string · classes	datetime string · date	iob float64	cob float64	ig float64
 30 values	 2017-06... 2017-07...	 -6.32 7.1k	 0 423	 -13.78 1.02k
apricot-waterfall-16	2017-06-23 00:00:00+00:00	1.772	24	116.672
apricot-waterfall-16	2017-06-23 00:00:01+00:00	4.141	2	148.403

Downloads last month 22,933

[Use this dataset](#)

Size of downloaded dataset files:
9.76 GB

Size of the auto-converted Parquet files:
9.76 GB

Number of rows:
825,857,015

CSTS: Correlation-based Benchmark in TS

- It systematically models known correlation structures between different time series variates
- It enables examination of how these structures are affected by **perturbation**
- It enables assessing **clustering validation** methods for correlation-based clustering

Data Perturbation Impact

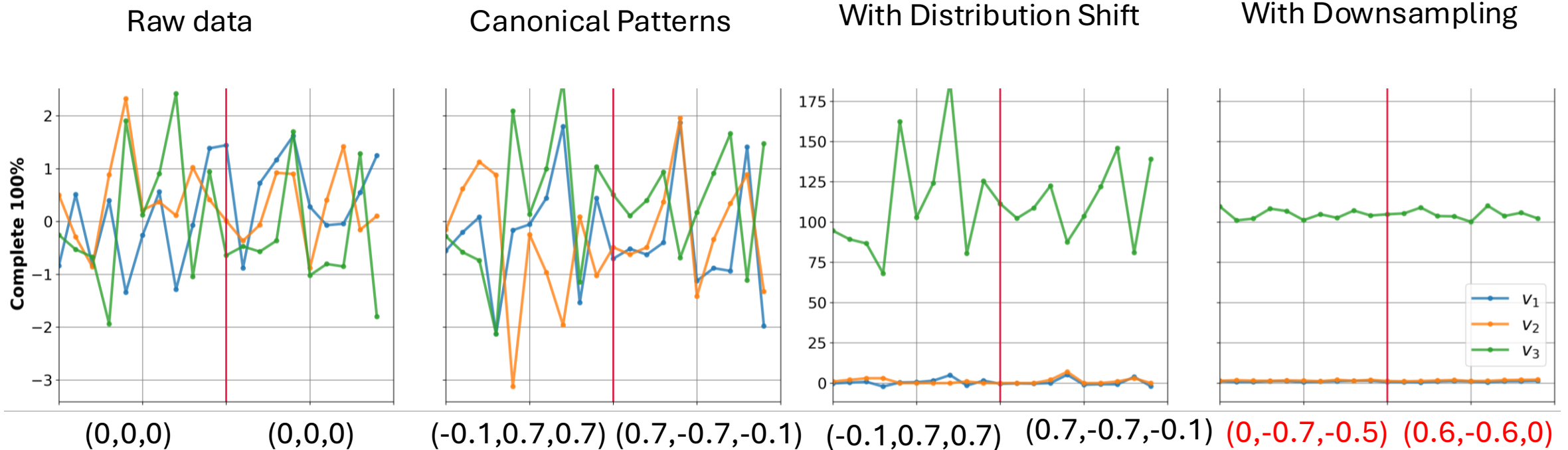
12 different variants to simulate various data conditions

- Set 1: Raw data following normal distribution (no correlation patterns)
- Set 2: Canonical patterns are encoded
- Set 3: Distribution shifts are introduced
- Set 4: Data is downsampled from 1 second -> 1 minute

All sets are tested on complete (100%), partial (70%), Sparse (10%).

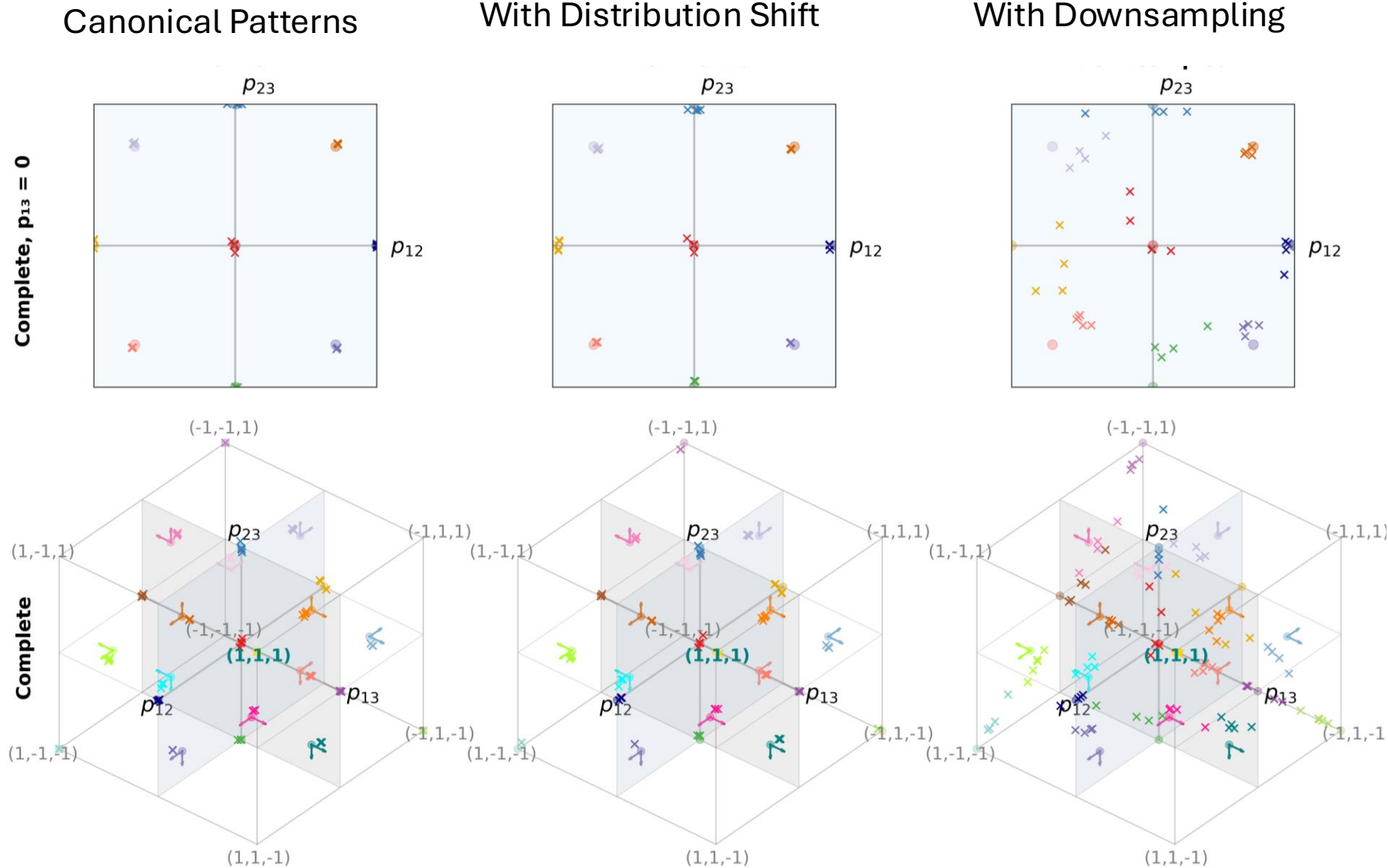
Data Perturbation

- Each panel shows the last 10 observations from two segments of one subject.
- Empirical correlation coefficients (a_{12}, a_{13}, a_{23}) shown for each segment; red lines indicate regime changes.



Perturbation Impact

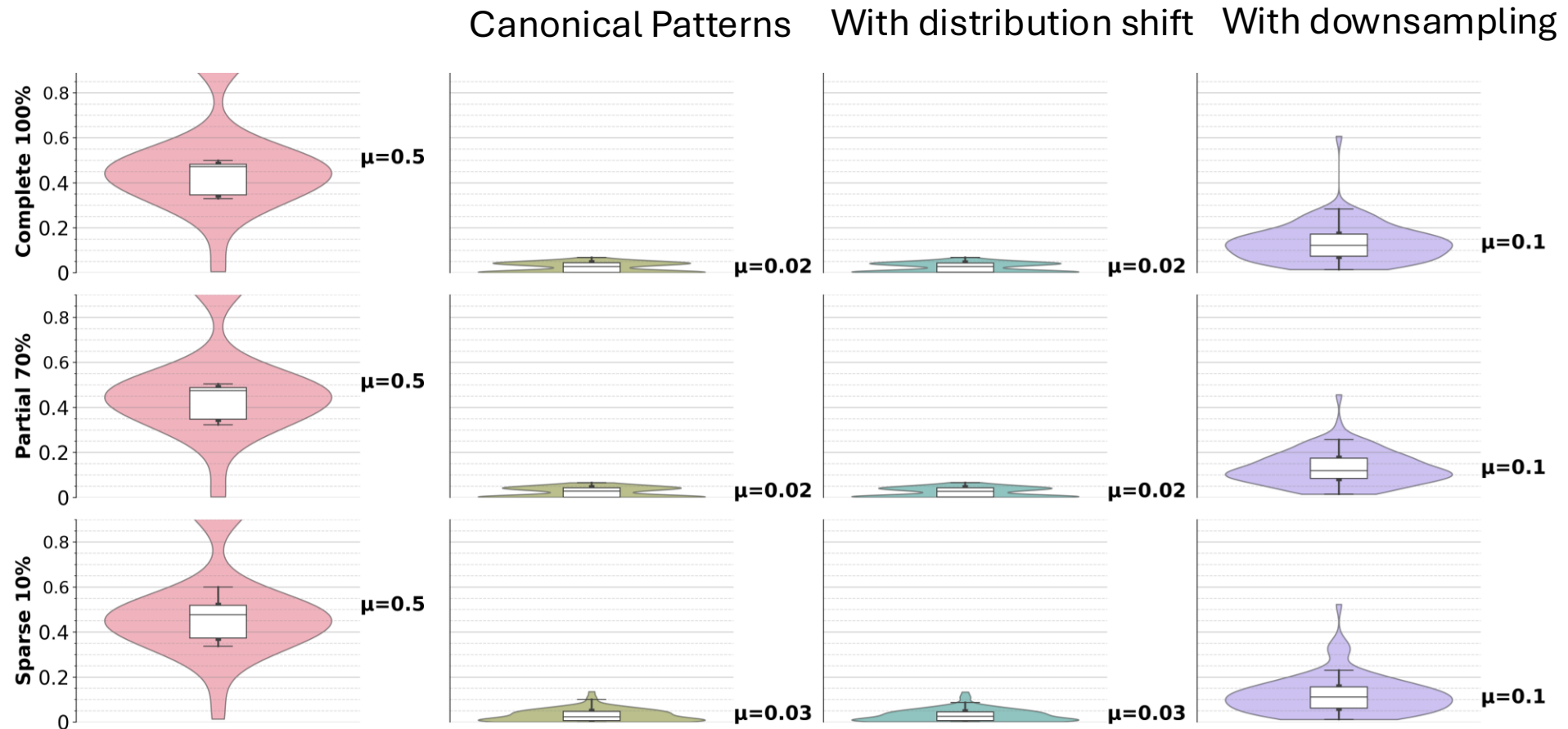
- **Downsampling** distorts correlation to moderate patterns
- **Distribution shifting** preserves patterns



The circles represent the relaxed canonical patterns P'_θ , the crosses their empirical versions A_m

Sparsification?

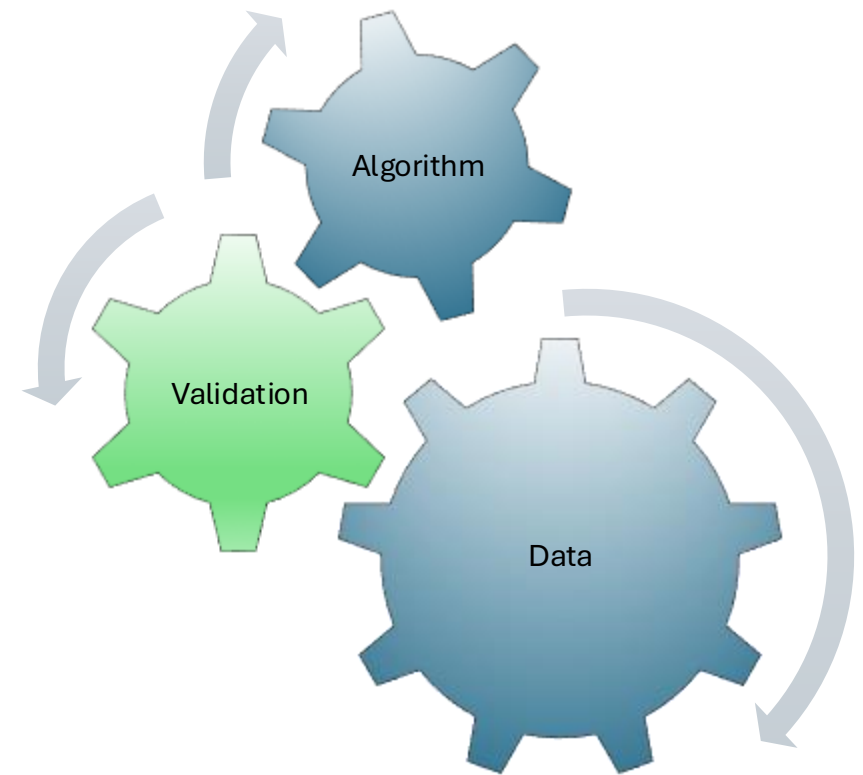
Sparsification shows unexpected correlation pattern resilience



Our finding suggests that correlation-based clustering can handle missing data, suggesting that algorithms should accommodate irregular sampling patterns.

Validation Methods in Correlation Space

- Internal validity indices have been pervasively developed using Euclidean data (usually L2 norm to measure similarity).
- Do they work for correlation matrices?
 - But correlation Elliptope space is different from Euclidean space

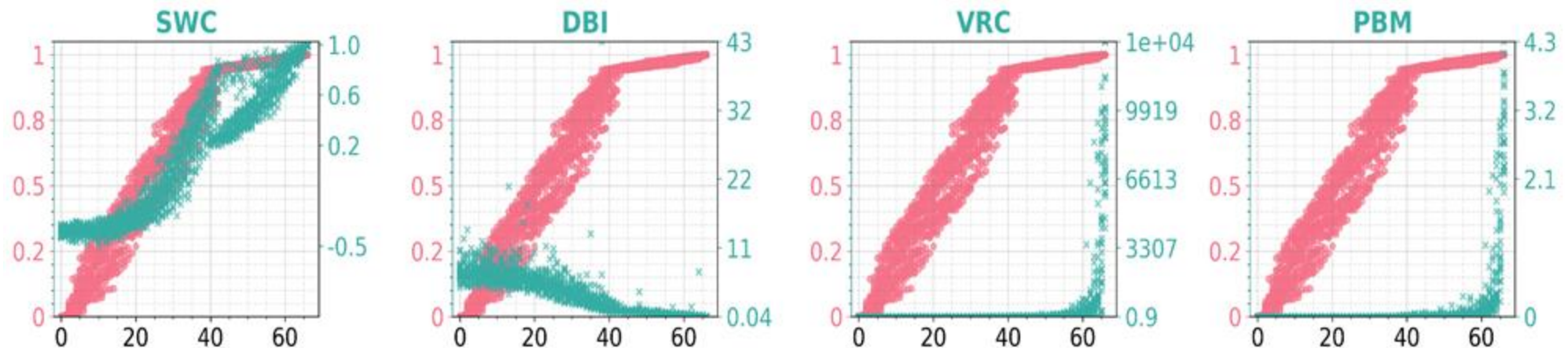


Validation Methods in Correlation Space

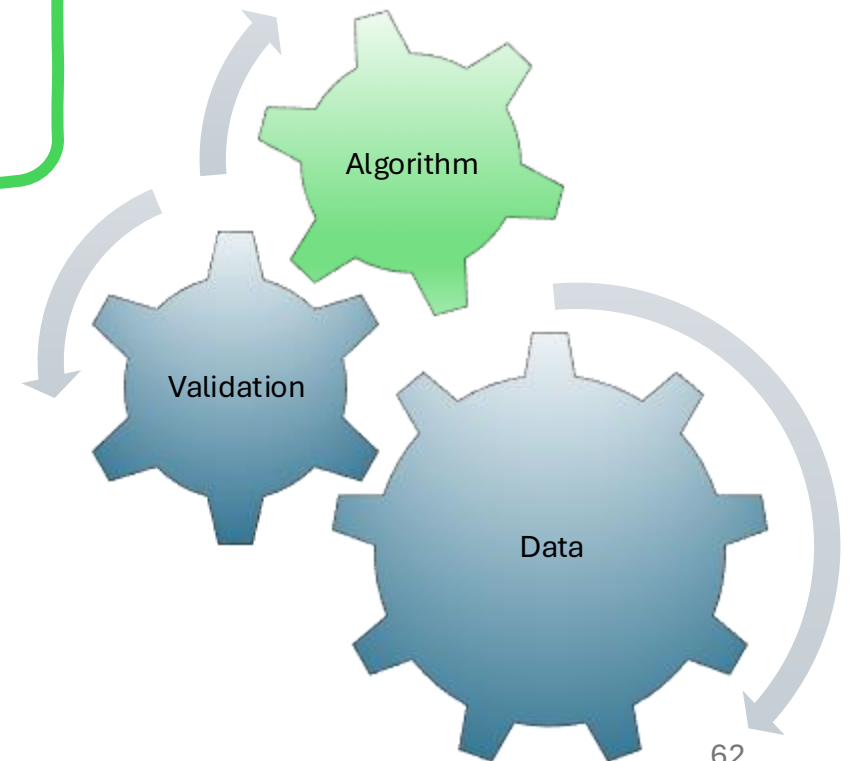
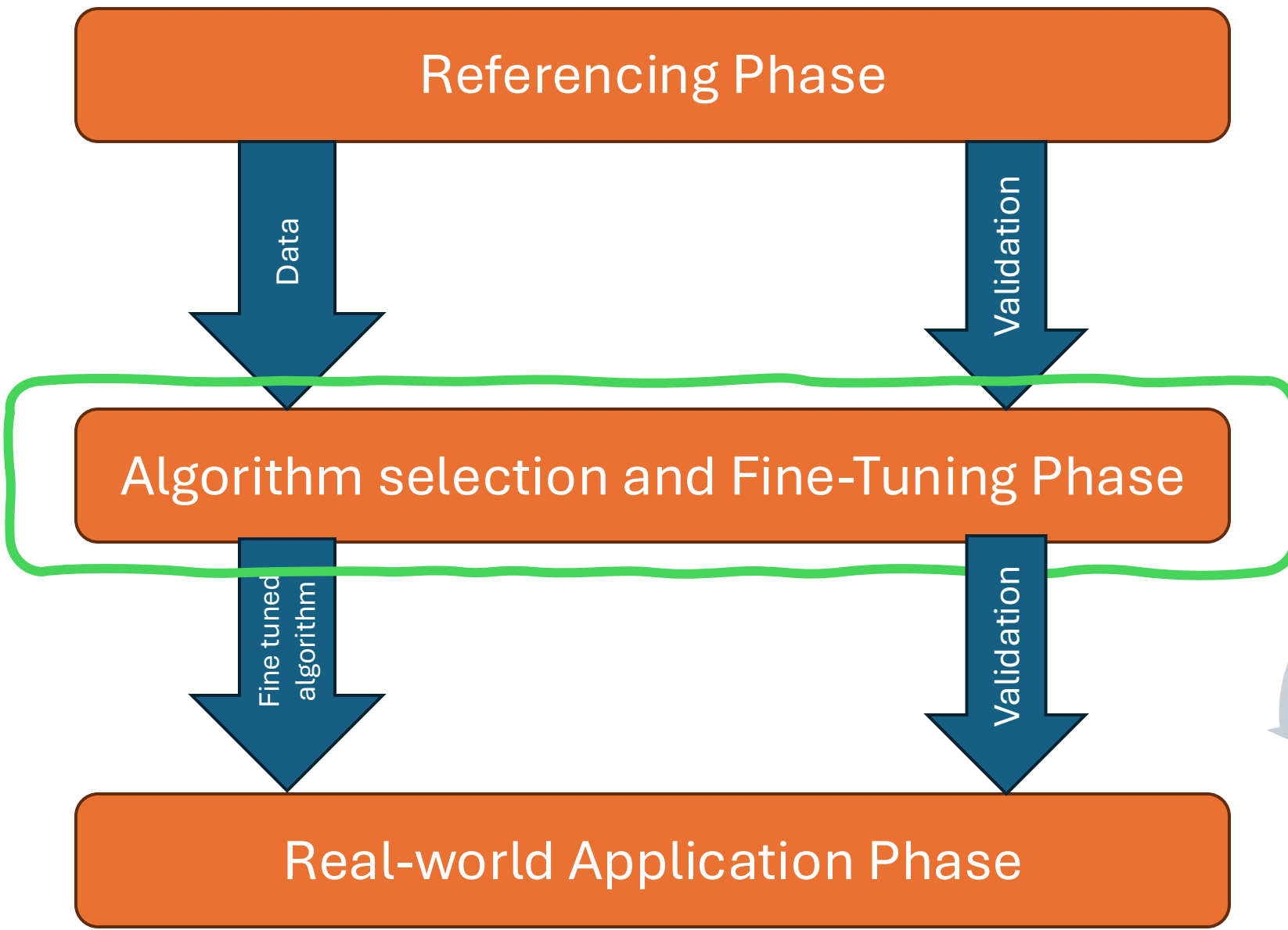
- We empirically validate existing clustering validation methods for correlation space applications
 - Distance functions
 - 15 distance functions
 - Internal validity indices
 - SWC Silhouette Width Criterion
 - PBM Pakhira-Bandyopadhyay-Maulik Index
 - VRC Calinski-Harabasz Index
 - DBI Davies Bouldin Index

Key Findings

Simple **Lp distances** outperform correlation-specific distances
SWC and **DBI** provide reliable quality assessment



Internal validity indices (green, x, right y-axes) compared to the Jaccard index (pink, o, left y-axes) of the 67 clusterings of various quality for 30 subjects. This is the data for the non-normal, partial (70%) variant.



Blueprint

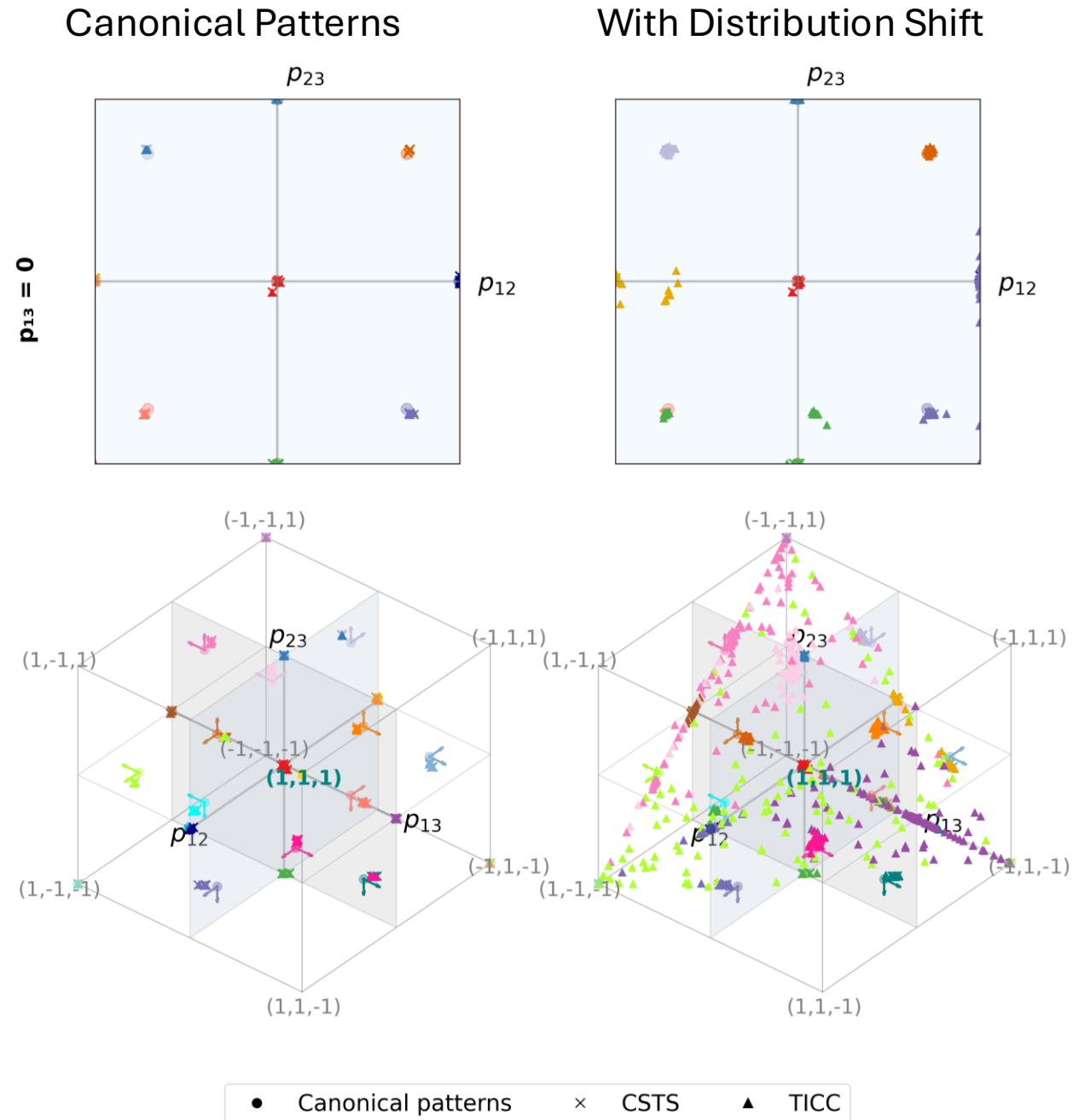
Real data to
cluster

- Use reference data to generate data with specific characteristics:
 - sampling rate, completeness, distribution, etc.
- Use generated data + recommended validation on Algorithm x:
 - Parameter tuning
 - Show strengths and pitfalls
- Apply the finetuned algorithm to your real data using the recommended parameters/validation strategy

Pattern discovery

How about TICC?

- These results show that TICC performs better for the complete data variant and **struggles more for sparsified data**
- TICC struggles with more complex correlation structures finds it easier to identify **simple correlation structures**
 - No correlation, all correlated, single strong correlation



Lessons Learnt / Future Directions

- Clustering should **not** be treated as approximate classification
- **Data structure** is central to meaningful results
- Objective evaluation helps enable **pattern discovery**
- Distinguishing artefacts from **data**, **algorithms**, and/or **validation** is essential for rigorous discovery.

Position: Clustering Results Are Unfalsifiable Without Structure-Specific Validation

Anonymous Authors¹

Abstract

Clustering discovers latent mathematical organisation in data but lacks principled criteria due to epistemic circularity. The discovered structure is used to justify the pipeline that discovered it, yet clustering is employed precisely because the structure is unknown. By applying four standard evaluation strategies to discover correlation structures, we demonstrate that current practices yield unfalsifiable and contradictory results without objective criteria to distinguish genuine patterns from artefacts. **This position paper argues that clustering evaluation requires objective, structure-specific validation to break the epistemic circularity that leads to unfalsifiability.** To break the circularity, we define optimal structure from the mathematical properties of each structure type and propose a validation framework using this definition as an independent reference standard. Instantiating the framework in a proof-

ganisation in the data). Throughout this paper, we refer to this mathematical organisation as *structure*. Clustering is ubiquitous and finds applications in preparing data for recommendation systems (Stergiopoulos et al., 2024), summarising complex embedding spaces (Xie et al., 2016), understanding biology (Xu & Tian, 2015), discovering patient subgroups (Gao et al., 2023), identifying financial market regimes (Marti et al., 2021), deciphering genomic patterns (Oyelade et al., 2016), revealing anomalies in control systems (Kiss et al., 2014), and evaluating AI systems (Rosa et al., 2023). Each clustering application requires choosing the structure(s) to be identified, preprocessing methods, algorithms, hyperparameters, quality measures, and interpretation approaches from numerous options (Hennig, 2016).

Current clustering evaluation practices produce contradictory and unfalsifiable results. No internal clustering validity index for assessing quality consistently outperforms others (Vendramin et al., 2010; Arbelaitz et al., 2013; Todeschini et al., 2024). Although these indices employ mathematical formulations that implicitly define what constitutes better